Who Gives Way? The Child Penalty within Spanish Couples

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Abstract

Who gives way when a couple has its first child? We study the child penalty within Spanish couples, using administrative records that identify both parents of each child and follow each parent's employment and labor income month by month. Where the father was the main earner, the share of couples in which he becomes the sole earner after the birth rises; where the mother earned at least half of the couple's income, it rises far less, and the gap between the two responses is about 20 percentage points. Earnings show the same asymmetry: when the mother contributed up to 60 percent of the couple's income, her earnings fall 25 to 30 percentage points more than his, whose earnings barely change. Up to that share, the mother adjusts more than the father whoever earned more, though by less where she had been the breadwinner. The estimates are preliminary.

Keywords: child penalty; gender gap; breadwinner; household decisions; Spain.

JEL codes: J13, J16, J22.

1 Introduction

Women's position in the labor market has improved for decades, but the pace of improvement has slowed (Olivetti et al., 2024). Over the same period, couples in advanced economies have come to rely increasingly on two incomes rather than one, although the second is often part-time (Lewis, 2001). Why gender gaps persist when both partners are in paid work is thus an open question.

Much of what remains appears when children arrive. After a first birth, mothers' employment and earnings fall sharply and do not recover, whereas fathers' barely move (Kleven et al., 2019b, 2025). Because these estimates are computed separately for women and men, they cannot say which partner gave up paid work in a given couple, or whether that depended on who earned more. Models of the household pose exactly that question, since they treat both partners' labor supply as the outcome of one joint decision (Lundberg and Pollak, 1996; Browning et al., 2014). This paper therefore takes the couple, not the parent, as the unit and asks how the cost of the first child is split between the two partners.

In the specialization model, a couple maximizes joint output by having the partner whose market wage is high relative to his or her productivity at home do the paid work (Becker, 1985, 1991). Since men earn more on average, the model assigns to the mother the extra home production that a child brings, and the child penalty becomes, for the most part, a consequence

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of the wage gap the couple already had. Bargaining models point the same way for a different reason: each partner would rather avoid the extra hours of care, and the one with the stronger position outside the marriage, typically the higher earner, is better placed to do so (Lundberg and Pollak, 1996).

Either way, the partner with the weaker market position should give way, so where the mother earned more, the father should absorb more of the cost. Nothing in this prediction requires mothers to be better carers. Long-run penalties are virtually the same for adoptive and biological mothers in Denmark (Kleven et al., 2021), and Norwegian adoptive and same-sex couples largely rule out both giving birth and the father's market advantage as causes of the penalty, leaving preferences, norms and discrimination (Andresen and Nix, 2022). Fathers' leave may nonetheless displace other care: in Spain, children eligible for four more weeks of paternity leave were more likely to show developmental delays, possibly because their fathers' care replaced maternal or formal childcare (Farré et al., 2024).

Gender norms lead to a different prediction. If people value acting in line with their gender identity, deviating from it is costly in itself (Akerlof and Kranton, 2000). A mother who keeps the larger paycheck and a father who cuts back on paid work both breach the prescription that men provide and women care (Bertrand et al., 2015; Bertrand, 2020), and couples may give up income to avoid such arrangements: in a laboratory experiment, partners specialized along gender lines beyond what their comparative advantage warranted (Couprie et al., 2020). On this account, the mother cuts back after the first birth whoever earned more, and the couple's response is asymmetric.

Studies that follow both partners show that relative earnings move the division of paid work only at the margin. A mother's higher relative earnings or education reduces her loss in the Netherlands (Artmann et al., 2022; Machado and Jaspers, 2023) and in Sweden, where parenthood even narrows the partners' income gap in the long run in about a fifth of couples, those in which the father's earnings prospects were weakest relative to the mother's (Angelov et al., 2016). The division itself does not reverse. Dutch mothers who had been the main provider still lose about a third of their earnings while their partners gain (Machado and Jaspers, 2023); in Germany and France the mother's loss is as large, or larger, where she had out-earned the father, and it is he who raises his earnings (de la Vega, 2022; Bonnet et al., 2025); and the employment of American mothers who had been the breadwinner falls less at first but converges with that of secondary earners ten years after the birth (Galván and García-Peñalosa, 2025). Dutch fathers' labor supply barely responds to the couple's relative earnings potential (Artmann et al., 2022), in the United States the long-run child penalty is about 50 percent even where the wife is much more educated than her husband (Cortés and Pan, 2023), and across European couples the mother's share of paid hours falls most where he was the main earner, but it also falls where she was (Bian et al., 2024). Almost all of this evidence is annual and concerns earnings. Closest to this paper, Querejeta (2026) follows Uruguayan couples month by month on the margin where specialization makes its sharpest prediction, which partner stops working: where the father had the higher earnings or education before the birth, more couples end up with the father as the only employed partner, but where the mother had them, such moves remain common and moves to the mother as the only employed partner do not take their place.

Evidence on norms points the same way. A norm that the husband should out-earn his wife can explain why couples in which she earns just over half are scarce (Bertrand et al., 2015), although a marriage market without such a norm can produce a similar scarcity (Grow and Van Bavel, 2020), and across European regions more egalitarian attitudes go with more couples in which both earn about the same, but not with more couples in which she out-earns him (Vitali and Arpino, 2016). Traditional gender beliefs held in adolescence predict larger child penalties in the United States (Rafols, 2025), and the woman's share of the couple's labor income falls

less after a first birth among couples socialized in East Germany than among couples socialized in the West (Jessen, 2022). In different-sex couples sex largely governs who specializes, and relative earnings matter little, while same-sex couples specialize considerably less (Siminski and Yetsenga, 2022; Moberg, 2016); in the United States, women who out-earn their male partners still do more housework than their partners, whereas in couples in which the man out-earns the woman, and in same-sex couples, the higher earner does less (Hancock et al., 2025).

Spain is a natural setting for this question. Mothers' long-run earnings loss after a first birth, about 28 percent, is similar to that in Sweden or the United States (de Quinto et al., 2021), the employment penalty is more than twice Portugal's (Kleven et al., 2025), gender convergence in the labor market has stalled since the early 2010s, with larger gaps among people with children (Hupkau and Ruiz-Valenzuela, 2022), and the successive extensions of paternity leave to sixteen weeks between 2017 and 2021, which fathers who took leave used nearly in full (Farré et al., 2025), did not detectably change the penalty (González et al., 2026). Evidence on how Spanish couples share that loss, however, is scarce. In a European survey panel for 2004 to 2008, Spanish women's share of the couple's earned income fell by about 3 percentage points after a birth, a fall that family benefits, counted as the woman's, largely offset (Dotti Sani, 2015). The two-week paternity leave introduced in 2007 raised mothers' employment and fathers' childcare and housework time only in couples with an intermediate gap in potential wages, proxied by the partners' age and education gaps (González and Zoabi, 2021). With administrative records from several sources, De Poli (2025) shows that a first child widens the employment gap between fathers and mothers by about 20 percentage points, a gap that, descriptively, narrowed by five points between 2016 and 2021 as paternity leave grew from two to sixteen weeks, but does not ask who earned more.

This paper asks whether the first birth affects couples differently when the mother out-earned the father before it, along two margins: which partner stops working, and how much labor income each partner loses. It draws on Spanish administrative records that identify both parents of each child and follow each parent's employment and labor income month by month, from two years before the birth through the first years after it; the estimates center on the year after the leave. Following Kleven et al. (2019a) and, for couples, Querejeta (2026), the design is an event study with the couple as the unit and the month as the period; as in González et al. (2026), who estimate individual child penalties on Spanish Social Security records, the reference month is nine months before the birth. It compares couples in which the mother is the same age, in the same calendar month, but who have their first child at different times.

The first birth reorganizes paid work within the couple. Six to eleven months after the birth, the share of couples in which both partners are employed falls by 18 percentage points, almost entirely replaced by couples in which only the father is employed, and the mother's share of the couple's labor income falls by 11 points.

The central finding is that this reallocation is asymmetric. Where the father was the breadwinner, the share of couples with the father as sole earner rises; where the mother was the breadwinner, earning at least half of the couple's income, the share with the mother as sole earner rises far less, if at all, and the gap between the two responses is 20 percentage points, 16 if the pre-birth trend is extended. Earnings tell the same story: relative to what each partner would have earned without a child, the mother loses 25 to 30 percentage points more than the father, whose earnings barely change, when she contributed up to 60 percent of the couple's income, while for the few couples in which she contributed more the data are inconclusive. Annual tax records, which allow a comparison with childless couples classified in the same way, are consistent with this: mothers who had out-earned the father do not lose a detectably different proportion of their earnings from other mothers.

Up to a 60 percent share, the mother adjusts more than the father whoever earned more,

though by less where she had been the breadwinner. This pattern fits gender norms better than a division of paid work by relative pre-birth earnings alone, although the design does not test norms directly. Section 5 compares regions with more and less traditional gender attitudes and finds no significantly larger asymmetry in the most traditional regions, a test with power only for large differences that cannot separate norms from regional economic conditions. In female same-sex couples, in which sex cannot decide who absorbs the birth, the point estimate for the mother who gave birth is close to that of mothers in different-sex couples, but 27 couples cannot tell the gap from zero.

This paper adds to this evidence in three ways. It asks, with Spanish administrative records that identify both parents of a child and follow each one month by month, whether the couple's response to the first birth depends on who earned more; the Spanish penalties so far describe mothers and fathers separately, and the couple-level evidence comes from survey data or does not ask who earned more. It tests the symmetry that an allocation by relative earnings alone predicts on the margin where that prediction is sharpest, which partner stops working, and finds, in a formal labor market in which paid paternity leave rose from four to twelve weeks over the births studied, the asymmetry documented for Uruguay (Querejeta, 2026): where the mother was the breadwinner, couples do not move to her as sole earner. It then asks how far the data can order the explanations: earnings tell the same story up to a 60 percent share, while attitudes across regions and female same-sex couples point the same way without the precision to settle the question. The result is a description of who gives way in Spanish couples, not an identification of why.

Section 2 describes the data, Section 3 the empirical strategy and Section 4 the results; Section 5 examines gender norms, across regions and in female same-sex couples, and Section 6 concludes.

2 Data

This study relies on a novel dataset of Spanish administrative records that can link a child to both the mother and the father, which allows the child penalty to be analyzed within the couple.

The dataset is the Spanish Household Panel (*Panel de Hogares*), built by the National Statistics Institute (INE), the Tax Agency (AEAT) and the Institute for Fiscal Studies (IEF) from administrative records for a sample of households stratified by region, household type and income, followed every year from 2016 to 2023, when it covers about 8 percent of the population of Spain outside the Basque Country and Navarre (Instituto de Estudios Fiscales, 2025). Crucially, the panel's kinship records identify both parents of a child, which makes the unit of analysis the couple: the two parents of their first child, whether or not they are married or later stay together.

The panel links its members to their Social Security records, prepared for the panel with the structure of the Continuous Sample of Working Histories (in Spanish, *Muestra Continua de Vidas Laborales*, MCVL), which allow each parent's monthly employment history to be constructed from before the panel starts up to December 2025. As in de Quinto et al. (2021), a parent's monthly labor income is the contribution base as an employee, and months without such a base enter with zero labor income. The base is summed over employers, excluding the bases on which the public employment service contributes for recipients of unemployment benefits, and the part above the statutory cap is imputed by Tobit models (Card et al., 2013), as Bonhomme and Hospido (2017) do with the MCVL. A parent is employed in a month when labor income is positive; parents on birth leave remain registered and keep their contribution

base, so they count as employed during the leave. A parent on unpaid childcare leave has no base and counts as not employed.

The sample is built in steps (Table 1), starting from the first child of both partners, a mother and a father, born in 2017–2022. Requiring both parents to live in the same panel household in the birth year removes 38 percent of the couples, mostly because one or both parents enter the panel only in a later year (Instituto de Estudios Fiscales, 2025). As in de Quinto et al. (2021) and González et al. (2026), the sample consists of employees: couples are dropped if, in the three years before the birth, a partner is self-employed or works as an employee with no base in the records of employee contribution bases, as agricultural and domestic employees do. This removes 39 percent of the couples left; after the birth, such months are missing rather than zero (5.4 percent of the main sample's couple-months from the birth month on). A further rule drops couples in which a partner earns a wage that the Social Security records miss, as civil servants with their own pension scheme do. Another keeps those in which both partners have some Social Security record between 25 and 60 months before the birth, similar to the requirement in Nylin et al. (2021) that both partners have earnings two years before the birth. The main sample is a balanced panel, as in Kleven et al. (2019a): the births of 2017–2020, each couple observed in every month from two years before to five years after the birth. It consists of 7,044 couples (Table 2), contributing 598,740 couple-month observations from January 2015 to December 2025. Estimates are weighted with the panel's household weight in the birth year. Between 24 and 13 months before the birth, both partners are employed in 80 percent of the couples.[1]

A check with annual data takes labor income, plus the parent's own birth and childcare benefit, from the panel's tax records, which include self-employment income.[2] Because the panel is new, Appendix B checks it against the Spanish child penalty estimated by de Quinto et al. (2021): with their design and the Social Security records linked to the panel's parents, the penalty ten years after the first birth is 0.262, against 0.282 in the original.

3 Empirical strategy

The effect of the first child is estimated with the event study of Kleven et al. (2019a), taking the couple as the unit and the month as the period, with the couple outcomes as in Querejeta (2026). For couple c in year-month s, τ_{cs} months after the first birth,

$$y_{cs} = \sum_{j \neq -9} \alpha_j \, \mathbf{1}[\tau_{cs} = j] + \sum_{k} \beta_k \, \mathbf{1}[a_{cs} = k] + \sum_{m} \gamma_m \, \mathbf{1}[s = m] + \varepsilon_{cs}, \tag{1}$$

[1] Labour Force Survey employment rates for Spain in 2018, 78 percent for women and 88 percent for men aged 25 to 49 living in a couple without resident children, imply that both partners were employed in 66 to 78 percent of such couples (Eurostat, 2026). Without the rule that both partners have some Social Security record 25 to 60 months before the birth, the panel gives 65 percent, just below that range. The two sources also differ in population and measurement: the survey covers all couples without resident children rather than couples about to have a first child, refers to one week, and counts self-employment and undeclared work, whereas here a partner is employed in a month with a positive contribution base as an employee, and the sample is restricted to couples of employees.

[2] The check compares 1,493 couples with a first birth in 2020–2021 with 3,212 childless married couples, classified in the same way by the woman's share of the couple's labor income and assigned placebo births (Kleven et al., 2019a), in a stacked triple difference estimated by Poisson pseudo-maximum likelihood (Cengiz et al., 2019; Chen and Roth, 2024). Mothers who had out-earned the father do not lose a detectably different proportion of their income from other mothers: over the year of the birth and the two following years, the difference (mothers who had out-earned the father minus other mothers) is −1.4 log points, with a 95 percent interval from −6.4 to +3.5. Measured from two years before the birth, the upper end of the interval is +6.0 log points, against an average loss of 10.9 for other mothers, so the data rule out only that these mothers lose a proportion more than about 55 percent smaller than other mothers (29 percent measured from the year before the birth); the bound is higher in restricted samples.

where y_{cs} is one of the couple outcomes defined below; $\mathbf{1}[\cdot]$ is the indicator function; j runs from -24 to $+60$ months relative to the birth, month 0 being the month of the birth, omitting the reference month -9; a_{cs} is the mother's age (calendar year minus year of birth), indexed by k; m indexes the year-months from January 2015 to December 2025; and ε_{cs} is the error term. The coefficient of interest is α_j for $j \geq 0$: the change in the outcome j months after the birth relative to month -9, net of the mother's age and of year-by-month effects. The couple outcomes are indicators of who is employed, both partners (dual earning), only the father (the father as sole earner), only the mother (the mother as sole earner) or neither, and the mother's share of the couple's monthly labor income, which is missing when the couple has none. A negative α_j for dual earning together with a positive one for the father as sole earner means that the birth moves couples from both partners employed to only the father employed, and a negative α_j for the mother's share means that her share of the couple's labor income falls.

Conditional on the mother's age and the year and month, event time varies because mothers have their first child at different ages, so the effects are identified by comparing couples in which the mother is the same age in the same month but has her first child at a different time (Kleven et al., 2019a); the assumption is that, without the birth, their outcomes would have evolved smoothly through the event, and the specification imposes that the effect at each horizon is the same across birth cohorts. Because every couple in the sample has a child, the specification omits couple fixed effects: with them, and without couples that are never treated, the event-time path would be identified only up to a linear trend (Borusyak et al., 2024). The equation is estimated by weighted least squares, with standard errors clustered by couple. Months 6 to 11 after the birth, once most leave has ended, give the main summary, and from month 47 on no couple in the sample is still childless, so estimates there rest on extrapolation across birth cohorts and are not interpreted.[3] A joint test that $\alpha_j = 0$ for $j = -24$ to -10 checks the months before the reference, and a sign is interpreted only if it survives extending the pre-birth trend, a line through zero at the reference month fitted by least squares to the coefficients of months -24 to -10; this is a linear extrapolation of the pre-birth drift, not the formal bounds of Rambachan and Roth (2023).

To ask whether the response depends on who earned more, couples are classified by the mother's share of the couple's labor income in months -36 to -25, before conception and before the months that test the pre-birth trend, so that any regression to the mean shows up in that test instead of being read as an effect (Daw and Hatfield, 2018). She is the breadwinner if her share is at least one half, as in about a third of couples; otherwise the father is. The sole-earner asymmetry compares two responses to the birth: the rise in the share of couples in which only the father is employed (the father as sole earner) among couples in which he was the breadwinner, and the rise in the share in which only the mother is employed (the mother as sole earner) among couples in which she was. It is the first rise minus the second. A rule that allocates paid work by relative market earnings alone, whatever the sex of the partner, predicts symmetry: for the same earnings advantage the two rises would be similar and the asymmetry close to zero. Prior earnings reflect wages, hours, occupation and earlier choices rather than comparative advantage at home, so the test asks whether the response violates that symmetry, not whether it rejects specialization in the strict sense. A positive asymmetry means that the share of couples with the father as sole earner rises when he was the breadwinner, but the share with the mother as sole earner does not rise to the same extent when she was, as gender norms predict. The two groups share no couples, so the variance of the difference is the sum of their variances.

Earnings are examined in the same way, following Machado and Jaspers (2023): equation (1),

[3] Upcoming waves of the Spanish Household Panel will allow this estimation, which future versions of this paper will incorporate.

with each parent's own age, is estimated for the mother's and the father's monthly labor income separately by the mother's share before conception, below 40 percent, 40 to 60 percent and above 60 percent. This exercise keeps the 3,550 couples in which both partners had positive labor income in each of the twelve months of that window, expresses each parent's effect as a percentage of the earnings predicted without the child (Kleven et al., 2019a), and computes the standard error of the gap between the two parents from both estimations, so that it accounts for their covariance. Specialization predicts that the gap between the mother's loss and the father's narrows or reverses where she contributed more than 60 percent; gender norms predict that it remains.

As in González et al. (2026), who estimate an individual-level version of this event study on monthly Social Security records (MCVL), the reference month is $\tau = -9$, around conception. With $\tau = -12$ instead, dual earning falls by 2.5 percentage points more and the share with the father as sole earner rises by 1.6 points more in every month after the birth, because both already move by that much in the three months up to conception. The share with the mother as sole earner, the mother's share, the sole-earner asymmetry and the father–mother gaps in earnings effects by her pre-birth share change by less than one point, and the last two by 1 to 6 points, without changing sign, if the pre-birth drifts are extended as in the figure notes.

4 Results

The first birth moves couples out of dual earning and toward the father as sole earner (Figure 1). Relative to month -9, the share of couples in which both partners are employed falls by 17.9 percentage points in months 6 to 11 after the birth (95 percent interval 15.5 to 20.3), and the share in which only the father is employed rises by 17.6 points (15.5 to 19.7); the share in which only the mother is employed does not change detectably (-0.8; -2.0 to 0.4). Before the reference month, a joint test does not reject that the dual-earning coefficients are zero ($p = 0.29$), and extending the pre-birth trend reduces the fall in dual earning to 15.8 points (12.1 to 19.6). Grouping couples by their arrangement at conception points the same way as Figure 1: among those in which both partners were employed, only the father is employed in 22 percent of months 6 to 11 after the birth, and only the mother in 3 percent (Appendix Figure 8).

The mother's share of the couple's monthly labor income falls by 10.7 percentage points in months 6 to 11 (9.4 to 12.1), with no detectable movement before the reference month ($p = 0.96$; Figure 2).[4] The fall is larger than the 3.4 points that Dotti Sani (2015) estimates for Spain in 2004–2008, with survey income of the twelve months before the interview, an indicator for children aged 0 or 1 and births of any order. In the annual tax records, the mother's own birth and childcare benefit in the year of the birth, 5,408 euros on average, is as large as the fall in her labor income relative to the year before, 5,443 euros, as a benefit that replaces her contribution base in full during the leave implies (Real Decreto-ley 6/2019, 2019). With the father's benefit and other transfers net of the fall in his labor income, the couple's annual income ends 4,064 euros above its level of that year, in nominal euros and including the income growth that would have occurred without a birth (Appendix Figure 9); these are accounting changes relative to the year before the birth, not effects of the birth.

[4]The births in the sample fall under four paternity leave regimes, four weeks from January 2017, five from July 2018, eight from April 2019 and twelve from January 2020, while maternity leave stayed at sixteen weeks. With a separate event-time path for each regime, a joint test does not reject that the fall in months 6 to 11 is the same under all four, for dual earning ($p = 0.97$) and for the mother's share ($p = 0.16$). Of the six differences from four weeks, only one excludes zero: the mother's share falls 4.1 percentage points less under eight weeks (0.3 to 8.0). Each regime is also a period of births, and months 6 to 11 under eight and twelve weeks overlap the COVID-19 pandemic, so these differences cannot be attributed to the leave. Comparing births three months before and after each extension, González et al. (2026) find no detectable change in the child penalty in earnings.

The response is asymmetric (Figure 3). Among couples in which the father was the breadwinner before conception, the share of couples with the father as sole earner rises by 18.6 percentage points in months 6 to 11 (15.9 to 21.4). Among couples in which the mother was the breadwinner, the share with the mother as sole earner rises far less, if at all (−1.4; −4.2 to 1.3). The sole-earner asymmetry, the difference between the two, is 20.1 points (16.2 to 24.0), and 15.6 points (9.1 to 22.1) when the pre-birth trend is extended: where she had been the breadwinner, the share with the mother as sole earner rises far less than the share with the father as sole earner where he had been. The raw transitions show the same pattern (Figure 4): between conception and months 6 to 11, couples with the father as sole earner go from 14 to 32 percent where the father was the breadwinner, while couples with the mother as sole earner go from 11 to 9 percent where the mother was. The asymmetry holds within 20 points of parity, with a breadwinner who earned more in two pre-birth years, among couples in which both were employed before conception, among births up to March 2019 and with missing months set against it, and it does not differ significantly across paternity leave regimes (Appendix F).

Up to a 60 percent share, earnings tell the same story (Figure 5). Relative to what each parent would have earned without the child, the mother's loss in months 6 to 11 exceeds the father's by 29.9 percentage points (23.4 to 36.4) where she had contributed less than 40 percent of the couple's labor income and by 25.2 points (20.6 to 29.8) where she had contributed 40 to 60 percent, while the father's own earnings barely change; both gaps survive extending the pre-birth trend. These gaps are those of the trough after the leave, not of the long run, and are not comparable with the 11.4 percent that de Quinto et al. (2021) find for the first year, an annual measure for the year of the birth (Appendix B). In the same records González et al. (2026) find a penalty of about 24 percent in months 6 to 11 (their Figure 7), with mothers about 18 percent below their reference and fathers about 5 percent above; here mothers lose more (23 to 31 percent) and fathers do not gain, in a sample in which both partners had positive labor income in each of the twelve months of the classification window. Where she had contributed more than 60 percent, the gap of 21.2 points (8.9 to 33.6) falls to 8.4 (−7.0 to 23.8) when the trend is extended, so for these 491 couples the data can neither confirm nor rule out that it persists. Compared with couples in which she had contributed less than 40 percent, the gap is 8.7 points smaller (−5.3 to 22.6), and 27.3 points smaller (9.5 to 45.2) when the trend is extended, so whether it narrows where she earned most depends on the pre-birth drift. Grouped instead by whether the mother was the breadwinner, as in Figure 3, the mother's loss in months 6 to 11 still exceeds the father's where she was, by 21.7 points, against 27.6 where he was, and by 11.3 against 29.2 when the trend is extended (Appendix Figure 10); the difference between the two gaps, 5.9 points (−2.0 to 13.8), becomes 17.8 (7.2 to 28.5) with the trend extended.

5 Gender norms

After the first birth, couples in which the father was the breadwinner before conception move toward the father as sole earner, but couples in which the mother was the breadwinner move far less, if at all, toward the mother as sole earner (Figure 3). That sole-earner asymmetry is what a norm that the husband should out-earn his wife predicts (Bertrand et al., 2015), but on its own it does not separate norms from other explanations. This section examines two further pieces of evidence: regions with more and less traditional gender attitudes, and female same-sex couples.

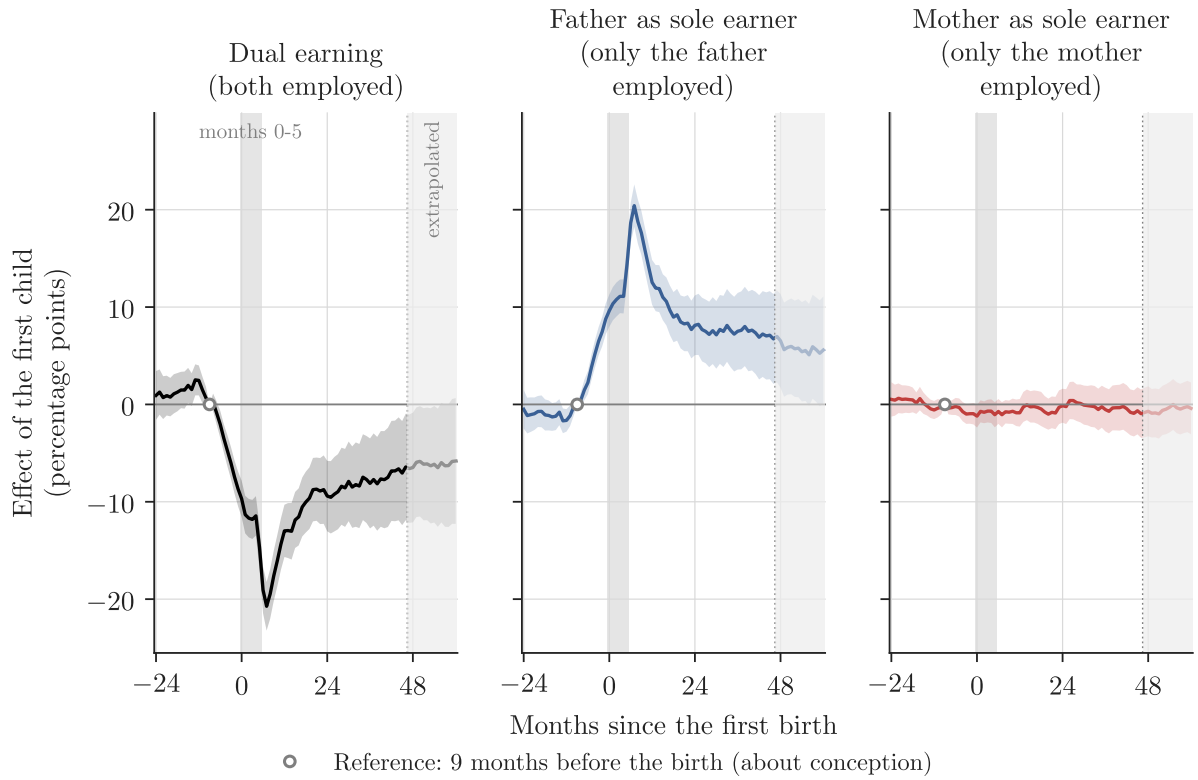


Figure 1: After the first birth, couples move from dual earning to the father as sole earner.

Notes: Each line is the estimated effect of the first child, relative to 9 months before the birth (about conception; hollow circle), net of the mother's age and of year-by-month dummies. Employed: a positive Social Security contribution base as an employee in the month; monthly labor income is that base. Shading around each line: 95 percent confidence interval. Vertical grey band: months 0 to 5, when most mothers, but not all, take their 16-week maternity leave; during the leave the mother keeps her contribution base. Shaded and faded, from month 47 on: these months fall in December 2020 or later, when every couple in the sample already had its first child; with no couple still waiting for its child to compare with, the estimate is extrapolated and not interpreted. Sample: 7,044 couples with a first child born between January 2017 and December 2020, neither partner self-employed, in agriculture or in domestic service in the three years before the birth, Spanish Household Panel linked to Social Security records, weighted. From the month of birth on, couple-months in which a partner is self-employed or an employee without a recorded contribution base are left out, not counted as zero: 2.3 percent of couple-months in months 6 to 11 and 8.4 percent from month 47 on. Between 12 and 9 months before the birth, dual earning already falls by 2.5 points and father as sole earner rises by 1.6, so the earlier months of those two lines sit away from zero. Without the child, in months 6 to 11, 82 percent of couples would be in dual earning, 10 percent would have the father as sole earner, 6 percent the mother as sole earner.

5.1 Traditional gender attitudes across regions

The test across regions, fixed before estimation, follows Casarico and Lattanzio (2023): they compare child penalties across Italian regions grouped by the share of residents who agree that a man's job is to earn money and a woman's job is to look after the home and family, and find the largest penalty in the most traditional third. The same item was fielded in Spain in 2012, before the births studied, in the Spanish edition of the International Social Survey Programme module on family and gender roles, with 2,595 interviews (Centro de Investigaciones Sociológicas, 2012). For each region, the index is the share of respondents who agree or strongly agree among those who answered; it is 9.7 percent in Madrid and 28.1 in Castile-La Mancha, against 18.3 nationally; the lowest, 8.5 in Extremadura, rests on only 59 answers (Appendix Figure 11). With 18 to 441 valid answers per region, sampling error accounts for much of the variation between regions,

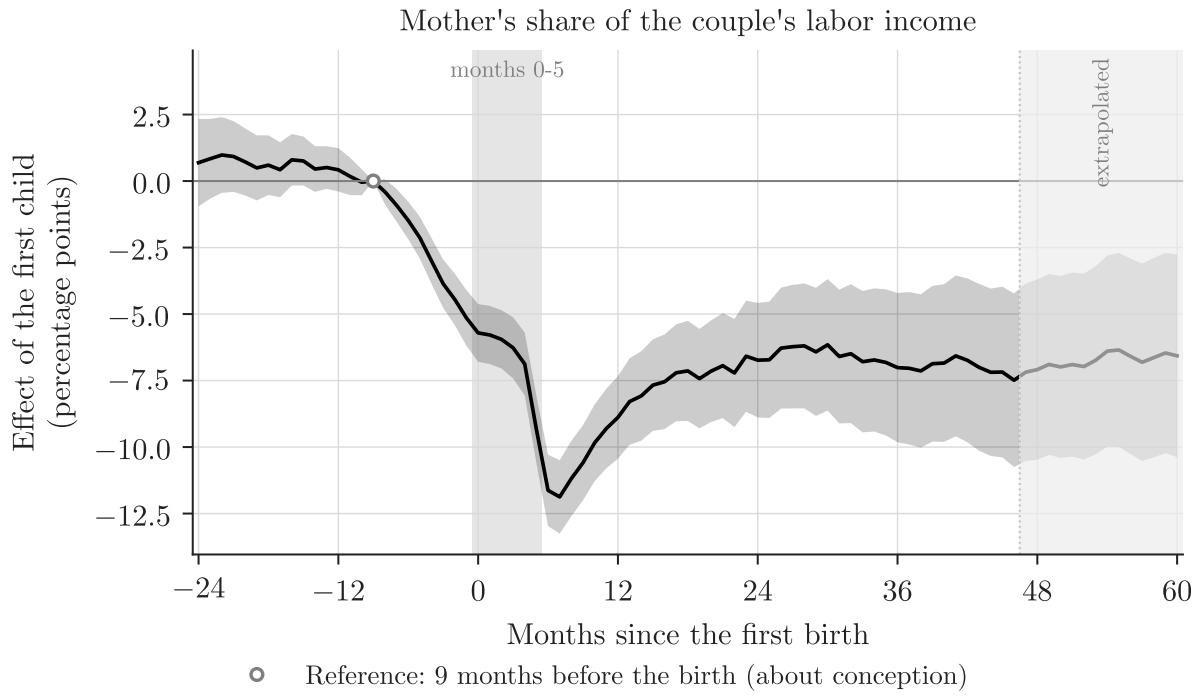


Figure 2: The mother's share of the couple's labor income falls after the first birth.

Notes: Each line is the estimated effect of the first child, relative to 9 months before the birth (about conception; hollow circle), net of the mother's age and of year-by-month dummies. Employed: a positive Social Security contribution base as an employee in the month; monthly labor income is that base. Shading around each line: 95 percent confidence interval. Vertical grey band: months 0 to 5, when most mothers, but not all, take their 16-week maternity leave; during the leave the mother keeps her contribution base. Shaded and faded, from month 47 on: these months fall in December 2020 or later, when every couple in the sample already had its first child; with no couple still waiting for its child to compare with, the estimate is extrapolated and not interpreted. Sample: 7,044 couples with a first child born between January 2017 and December 2020, neither partner self-employed, in agriculture or in domestic service in the three years before the birth, Spanish Household Panel linked to Social Security records, weighted. From the month of birth on, couple-months in which a partner is self-employed or an employee without a recorded contribution base are left out, not counted as zero: 2.3 percent of couple-months in months 6 to 11 and 8.4 percent from month 47 on. Outcome: the mother's monthly labor income as a share of the couple's, in percentage points; months in which neither partner is employed are left out. Without the child, in months 6 to 11, the mother's share would be 44 percent.

most of all in the middle of the ranking, which biases comparisons between groups of regions toward zero.

Couples are assigned the region where they lived in the year of the birth, and the 15 regions in the panel are split into thirds of five: the most traditional third (Asturias, Castile-La Mancha, Galicia, La Rioja and Valencia; 1,626 couples) is compared with the other ten (5,270 couples), and Ceuta and Melilla, with no interviews, are left out (148 couples). Equation (1) is estimated separately in each group, and two contrasts are read in months 6 to 11 after the birth: the sole-earner asymmetry defined in Section 3 and the fall in dual earning. Gender norms predict both to be larger in the most traditional third. A difference counts only if its 95 percent interval excludes zero and its sign survives three checks: extending the pre-birth trend (Rambachan and Roth, 2023), grouping regions by an index that averages three standardized items, as Kleven

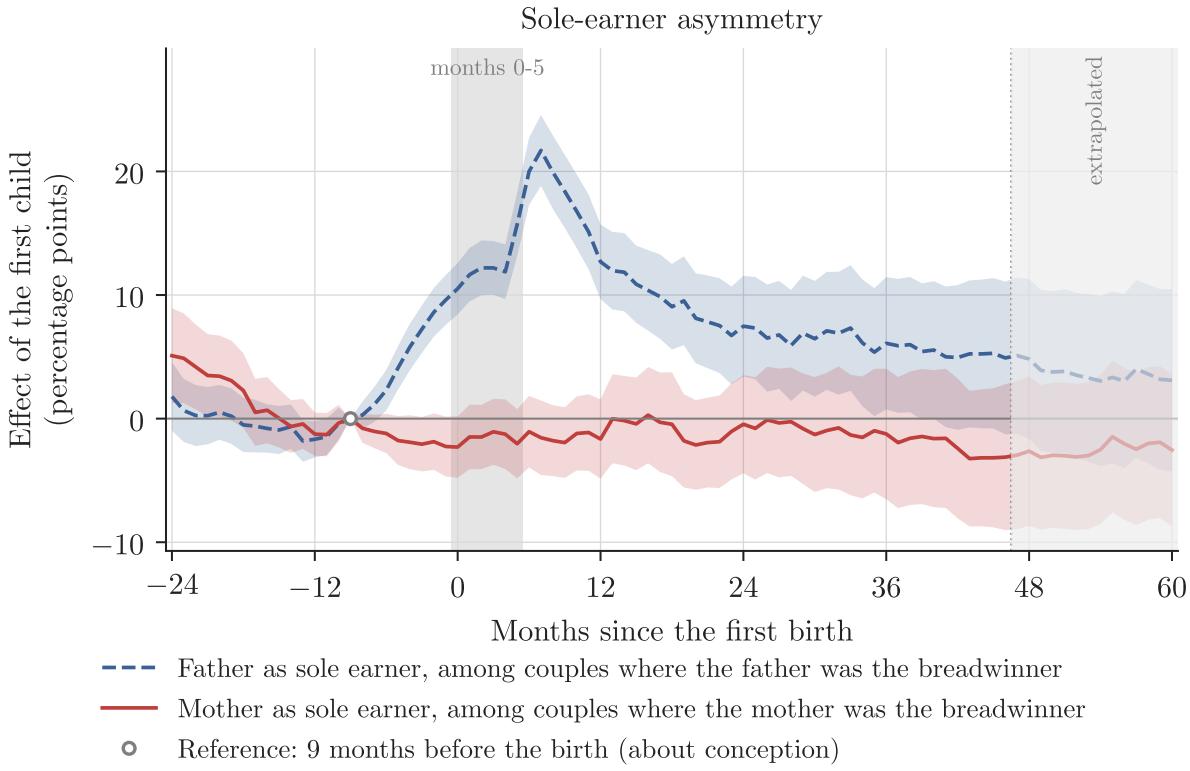


Figure 3: Six to eleven months after the birth, couples move to the father as sole earner when he was the breadwinner, and far less, if at all, to the mother as sole earner when she was.

Notes: Each line is the estimated effect of the first child, relative to 9 months before the birth (about conception; hollow circle), net of the mother's age and of year-by-month dummies. Employed: a positive Social Security contribution base as an employee in the month; monthly labor income is that base. Shading around each line: 95 percent confidence interval. Vertical grey band: months 0 to 5, when most mothers, but not all, take their 16-week maternity leave; during the leave the mother keeps her contribution base. Shaded and faded, from month 47 on: these months fall in December 2020 or later, when every couple in the sample already had its first child; with no couple still waiting for its child to compare with, the estimate is extrapolated and not interpreted. Sample: 7,044 couples with a first child born between January 2017 and December 2020, neither partner self-employed, in agriculture or in domestic service in the three years before the birth, Spanish Household Panel linked to Social Security records, weighted. The breadwinner is the mother if she earned at least half of the couple's labor income 25 to 36 months before the birth, and the father otherwise (4,396 couples where the father was and 2,524 where the mother was, 6,920 in all; the 124 with no labor income then are left out). From the month of birth on, couple-months in which a partner is self-employed or an employee without a recorded contribution base are left out, not counted as zero: 2.3 percent of couple-months in months 6 to 11 and 8.2 percent from month 47 on. Sole-earner asymmetry: the rise in the share of couples in which only the father is employed, among couples in which the father was the breadwinner, minus the rise in the share in which only the mother is employed, among couples in which the mother was the breadwinner; it is close to zero if couples reallocate paid work only according to who earned more. From 24 to 12 months before the birth both lines drift down, as expected from regression to the mean, since the groups are selected on that earlier period. Between 12 and 9 months before the birth both lines rise again, by 1 to 2 points. In months 6 to 11 the asymmetry is 20.1 points (95 percent interval 16.2 to 24.0), and 15.6 (9.1 to 22.1) if each line's linear pre-birth trend, fitted on months −24 to −10 through zero at the reference month, continued after it. Without the child, in months 6 to 11, the father would be the sole earner in 14 percent of couples where he was the breadwinner, the mother in 11 percent of couples where she was.

(forthcoming) does,[5] and leaving out one region at a time (Cameron and Miller, 2015).

[5]The three items are that a man's job is to earn money and a woman's job is to look after the home and family, the item of the main index; that a preschool child is likely to suffer if the mother works; and, reversed, that a working mother can establish just as warm and secure a relationship with her children as a mother who does not work. Items are scored so that higher is more traditional. The last two are Kleven's; his first item, that the man should be the achiever outside the home, is not in the 2012 Spanish survey.

The asymmetry is not significantly larger in the most traditional regions: 24.2 percentage points, against 18.7 in the other regions, a difference of 5.5 points (95 percent interval −4.1 to 15.1) that falls to 0.6 without Galicia (Appendix Table 6). The interval rules out differences larger than about 15 points, not smaller ones; before estimation, the design had 80 percent power only for differences of about 13 points or more. Dual earning falls by 23.0 points in the most traditional regions and by 16.0 in the others, a fall 6.9 points larger (a difference of −6.9 in Appendix Table 6, 95 percent interval −12.6 to −1.3) whose sign holds in the three checks, although it is not significant with the pre-birth trend extended (−5.3) or with the three-item index (−4.2). Standard errors are clustered by couple, which treats the 15 regions as fixed, so intervals with regions as the unit would be wider. By the rule above, the larger fall in dual earning in the most traditional regions is a secondary result; the asymmetry, the main contrast, is not significantly larger.[6]

Two descriptive extensions, added after this test and fixed before being estimated, show no steady gradient. By thirds, as in Figure 7 of Casarico and Lattanzio (2023), in months 6 to 11 the asymmetry is 19.1, 18.6 and 24.2 points from the least to the most traditional third and the fall in dual earning 17.7, 15.1 and 23.0 (Figure 12); in months 24 to 35 the point estimate of the asymmetry, without intervals, is largest in the least traditional third. Without grouping, regressing each region's asymmetry in months 6 to 11 on the index, as Kleven (forthcoming) regresses state penalties on a norms index but without his state fixed effects, gives −1.5 points of asymmetry per 10 points of agreement across the 15 regions (unweighted; 95 percent interval −6.7 to 3.7), where norms predict a positive slope.

The pattern by thirds in months 6 to 11 resembles the Italian one, but it is not robust, and, as Kleven (forthcoming) notes for US states, a single cross-section of regions cannot separate norms from everything else that differs between them. Before the birth, dual earning was less common in the most traditional third, mothers there were less often employed, and couple labor income was 17 percent lower. Kleven turns to changes within states over five decades, which he still calls correlational, and to people who move; a single survey year and the few parents born in another region allow neither. The item also measures attitudes in the adult population, not in the couples studied, although in the United States children's gender-role attitudes are correlated with their mothers' (Farré and Vella, 2013). Men agree more often than women: 22 and 15 percent of those who answered agree or strongly agree.

5.2 Female same-sex couples

In female same-sex couples both partners are women, so a norm about which sex should earn cannot decide which of them absorbs the birth, which leaves giving birth and prior labor income to compare (Andresen and Nix, 2022). As in the annual check described in Section 2, income comes from the panel's annual tax records: each partner's labor income plus own birth and childcare benefit, by year t relative to the birth. Neither the tax records nor the Social Security records show which partner gave birth, but the weeks of birth benefit paid to each partner reveal it. For a single birth in 2018–2020, the mother who gave birth was entitled to 16 paid weeks of leave and the other parent, the other mother in female same-sex couples, to at most 15, own leave plus any weeks transferred by the mother who gave birth (Farré et al., 2025; Real Decreto-ley 6/2019, 2019). The mother who gave birth is therefore identified as the partner

[6]The same split, applied after the employment results to the mother's share of the couple's labor income, gives a fall in months 6 to 11 of 12.2 percentage points in the most traditional regions and 10.2 in the others; the difference, −2.0 points (95 percent interval −5.3 to 1.3), that is, a fall 2.0 points larger in the most traditional regions, has the sign of the result for dual earning, on which it partly depends by construction, but is not significant (it is, at 5 percent, with the three-item index, which is not the main index) and rules out a fall more than 5.3 points larger (Figure 6).

paid at least 15.5 weeks of birth benefit, measured with her contribution base of the month before the birth, when the other partner was paid fewer. The rule classifies only couples in which the mother who gave birth was an employee, or on unemployment benefit, in the month before the birth and took her full leave; among different-sex couples so classified, it picks the mother in 98.7 percent of cases. Following the within-couple comparison of Andresen and Nix (2022), but in log points estimated by Poisson pseudo-maximum likelihood instead of in levels as a percentage of predicted income, the gap is the change since $t = -1$ in this income for the mother who gave birth, or for the lower earner, minus the change for her partner, averaged over $t = 0$ to 2, in log points. In 27 female same-sex couples, the gap for the mother who gave birth is −12.5 log points (95 percent interval −26.8 to 1.7; Appendix Figure 13), against −13.2 for mothers in different-sex couples classified the same way; the female same-sex gap is concentrated in the year after the birth, and two years before the birth it was +0.6 (s.e. 13.6, 17 couples), too imprecise to rule out a pre-birth gap. Holding fixed the difference between the two partners' employee contribution bases in $t = -4$ and $t = -3$, as Andresen and Nix (2022) do with earnings, it barely changes: −13.4 (−27.2 to 0.3). Partners can also be ranked by prior labor income, which covers 83 female same-sex couples with a first birth in 2017–2022: the comparison adds couples with a first birth in 2022, whose income the tax records show up to $t = +1$, and ranks partners by their labor income in the tax records whenever both have a tax record in $t = -4$ or $t = -3$, which also ranks partners whose labor income falls outside employee contribution bases, and by their employee contribution bases otherwise. In different-sex couples ranked this way, the mother falls behind the father whether she earned less before the birth (−10.9, 95 percent interval −12.5 to −9.3) or more (−18.5, −20.3 to −16.7); in female same-sex couples, the partner who earned less is estimated to gain 15.3 log points on the other (−2.4 to 33.0; Appendix Figure 14), an estimate that depends on how the partners are ranked. The point estimates are compatible with the birth rather than prior labor income driving the gap, as in 914 Norwegian female same-sex couples, where birth mothers had earned more than co-mothers on average and still lost 14 percent of their predicted earnings the year after the birth, against 4 percent, a difference no longer significant five years after the birth (Andresen and Nix, 2022). In Norway the gap between the partners of female same-sex couples, 10 percent, is about half the 20 to 25 percent penalty of mothers in different-sex couples (Andresen and Nix, 2022); with 27 couples, the interval (−26.8 to 1.7) includes that gap, the −13.2 of mothers in different-sex couples, and no gap. These samples, however, cannot tell the two apart: both intervals include zero, and the gap by prior labor income had already moved between $t = -2$ and $t = -1$ as much as after the birth. The 20 male same-sex couples that can be ranked in the same way are too few to estimate this gap reliably.

Employment points the same way. Among couples in which the mother who gave birth, identified in the same way, was employed in the month before the birth, her employment falls in months 6 to 11 by 19.1 percentage points (95 percent interval 7.7 to 30.5) in 47 female same-sex couples, against 17.8 for mothers and 2.1 for fathers in different-sex couples (Figure 7). The other mother's employment barely moves relative to month −9 (−1.9) but falls by 11.0 points relative to month −12, so the data cannot say whether co-mothers also cut back, as Andresen and Nix (2022) find in Norway outside the leave.

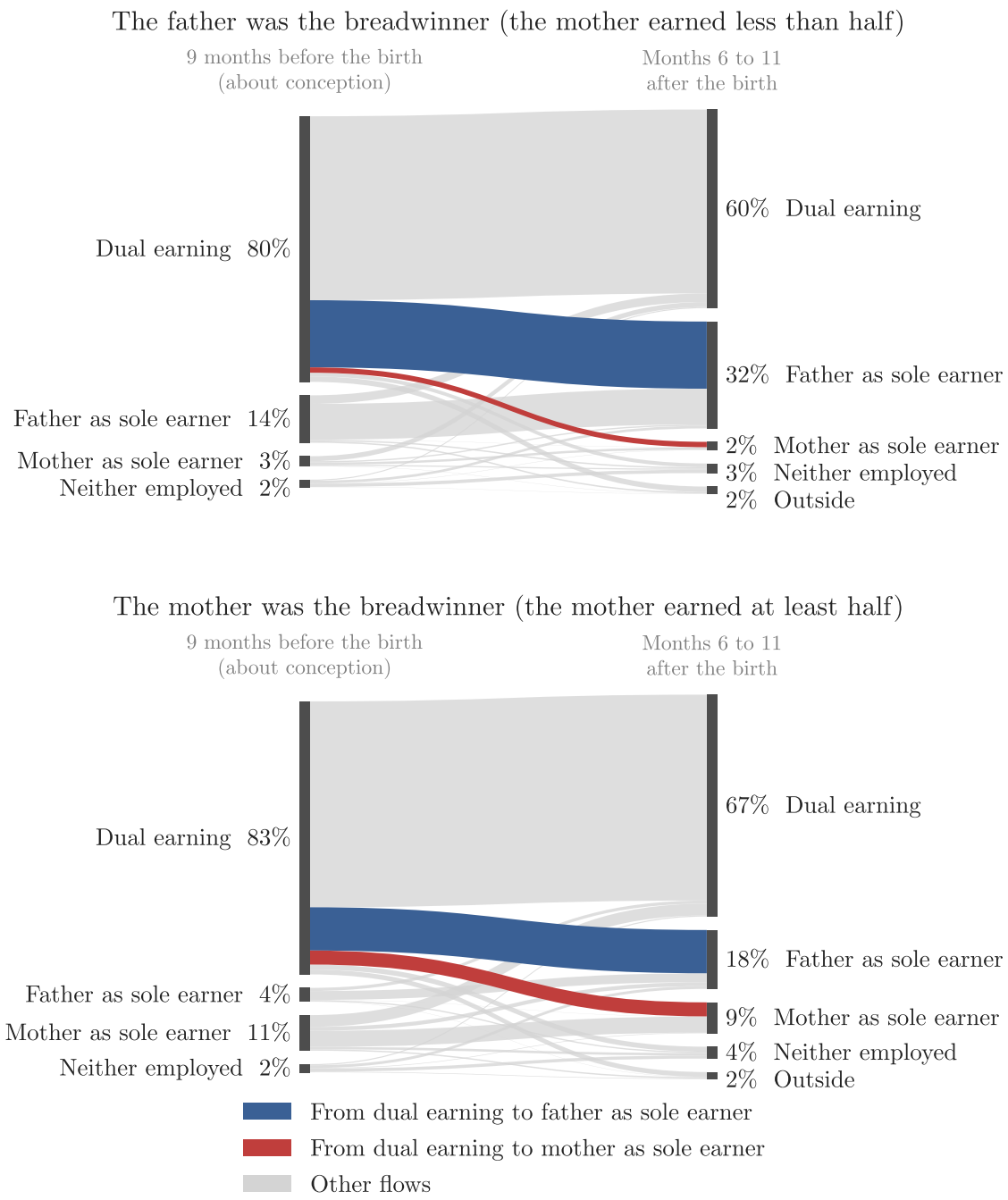


Figure 4: Even when the mother was the breadwinner, couples in which both were employed move after the birth to the father as sole earner far more often than to the mother as sole earner.

Notes: Descriptive: raw transitions, not net of what would have happened without the child; Figure 3 gives the estimated sole-earner asymmetry. Left column: couples by employment arrangement 9 months before the birth (about conception). Right column: their months 6 to 11 after the birth, by arrangement; each couple's six months are split across the arrangements it was in, as in Figure 8. Percentages and band widths: weighted percent of couples, 9 months before the birth on the left and on average over months 6 to 11 on the right. Where the father was the breadwinner, couples with the father as sole earner go from 14 to 32 percent; where the mother was, couples with the mother as sole earner go from 11 to 9 percent. Highlighted bands: of the months 6 to 11 of couples in which both were employed at conception, 25 percent are spent with the father as sole earner and 2 percent with the mother as sole earner when the father was the breadwinner, and 16 percent and 5 percent when the mother was. Employed: a positive Social Security contribution base as an employee in the month; birth leave counts as employed and unpaid childcare leave as not employed. Outside: months in which a partner is self-employed or an employee without a recorded contribution base. Sample: 7,044 couples with a first child born between January 2017 and December 2020, neither partner self-employed, in agriculture or in domestic service in the three years before the birth, Spanish Household Panel linked to Social Security records, weighted. The breadwinner is the mother if she earned at least half of the couple's labor income 25 to 36 months before the birth, and the father otherwise (4,396 couples where the father was and 2,524 where the mother was, 6,920 in all; the 124 with no labor income then are left out).

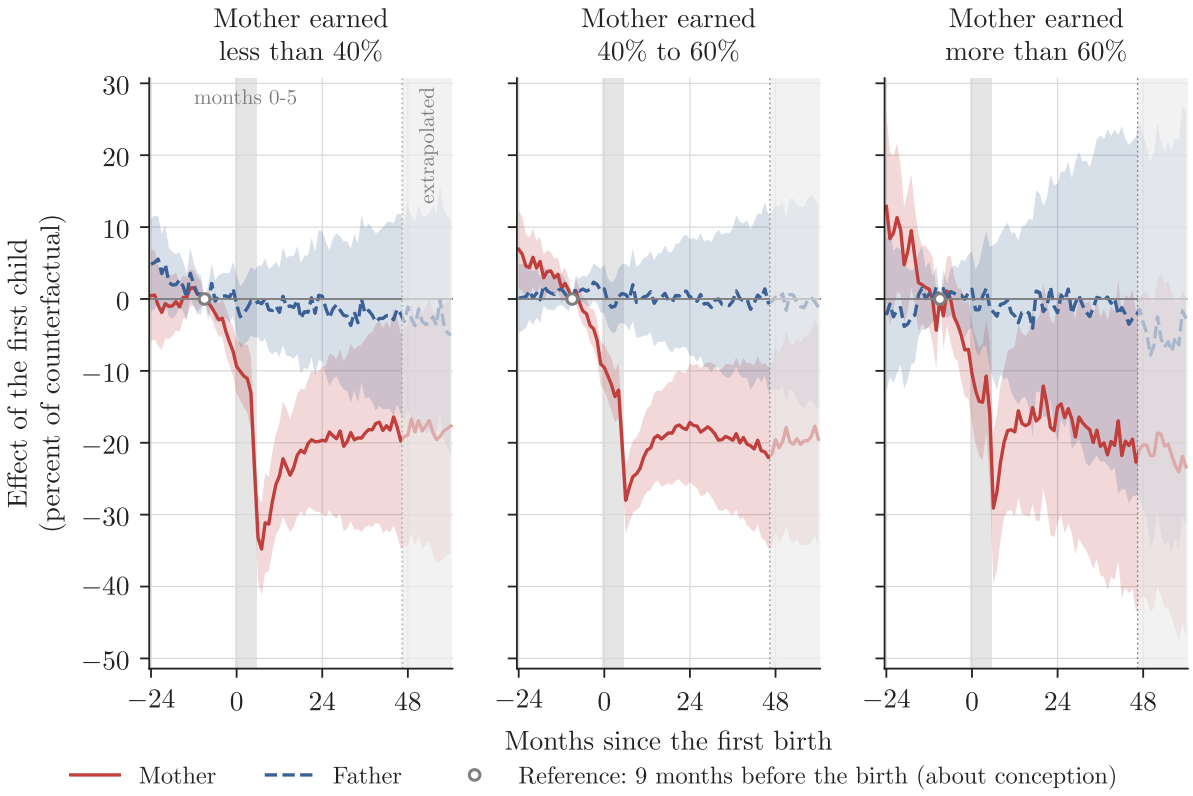


Figure 5: Six to eleven months after the first birth, the mother's labor income falls far more than the father's when she earned up to 60 percent of the couple's labor income; above 60 percent, the data cannot tell.

Notes: Each line is the estimated effect of the first child on that parent's monthly labor income, net of that parent's age and of year-by-month dummies, as a percentage of what the model predicts without the child, relative to 9 months before the birth (about conception; hollow circle). Employed: a positive Social Security contribution base as an employee in the month; monthly labor income is that base. Shading around each line: 95 percent confidence interval. Vertical grey band: months 0 to 5, when most mothers, but not all, take their 16-week maternity leave; during the leave the mother keeps her contribution base. Shaded and faded, from month 47 on: these months fall in December 2020 or later, when every couple in the sample already had its first child; with no couple still waiting for its child to compare with, the estimate is extrapolated and not interpreted. Panels: the mother's share of the couple's labor income 25 to 36 months before the birth (1,191, 1,868 and 491 couples). From 24 months before the birth to the reference month the mother's line goes from +13 percent to 0 in the right panel and from +7 percent to 0 in the middle one, and in the left one the father's line goes from +5 percent to 0; if each line's drift went on after the reference month, the gap between the father's and the mother's lines in months 6 to 11 would be 8 points instead of 21 in the right panel, 19 instead of 25 in the middle one and 36 instead of 30 in the left one. Without the child, in months 6 to 11, mothers would earn 1,680, 2,405 and 4,074 euros a month and fathers 3,480, 2,489 and 2,036. Sample: 3,550 couples with a first child born between January 2017 and December 2020 in which both partners were employed in each of those 12 months; neither self-employed, in agriculture or in domestic service in the three years before the birth; Spanish Household Panel linked to Social Security records; weighted. From the month of birth on, couple-months in which a partner is self-employed or an employee without a recorded contribution base are left out, not counted as zero: 1.6 percent of couple-months in months 6 to 11 and 6.9 percent from month 47 on. Panels as in Machado and Jaspers (2023, figure 2); unlike them, the coefficients on the age and year-by-month dummies are estimated within each panel.

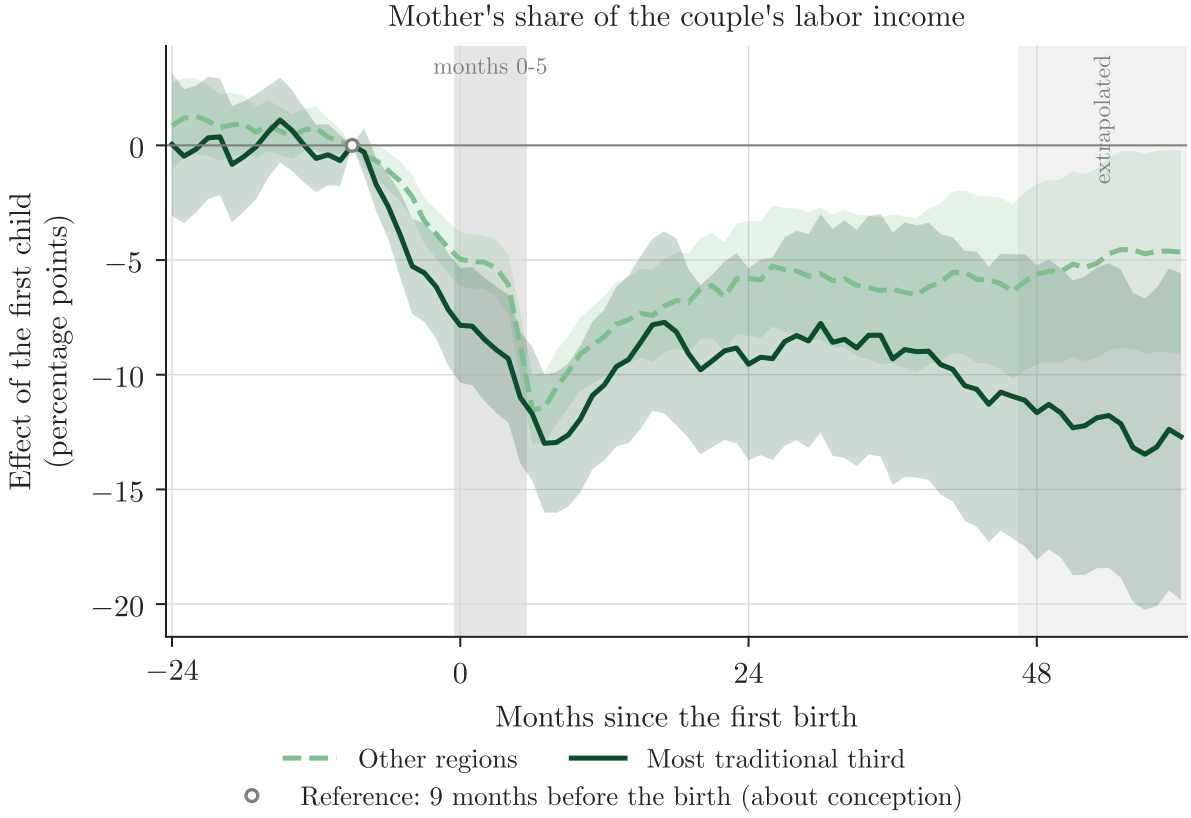


Figure 6: In months 6 to 11 after the first birth, the mother's share of the couple's labor income does not fall significantly more in the most traditional regions.

Notes: Lines: effects of the first child on the mother's share of the couple's labor income, in percentage points, relative to 9 months before the birth (hollow circle), estimated by weighted least squares separately in the most traditional third of the 15 regions (1,622 couples) and in the other ten regions (5,257), grouped as in Section 5.1 and listed under Figure 11; dark green, most traditional third; light green, the other regions. Months in which the couple's labor income is zero or not fully recorded are left out. Bands: 95 percent confidence intervals. Grey band: months 0 to 5; light grey band, from month 47: extrapolated. Months 6 to 11, with the checks of the test: row 5 of Table 6. In months −24 to −10, the coefficients are not jointly different from zero in either group ($p = 0.646$ and $p = 0.895$). Descriptive, added after the employment results; the share partly depends on dual earning by construction. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF), the linked Social Security records (MCVL) and CIS study 2942.

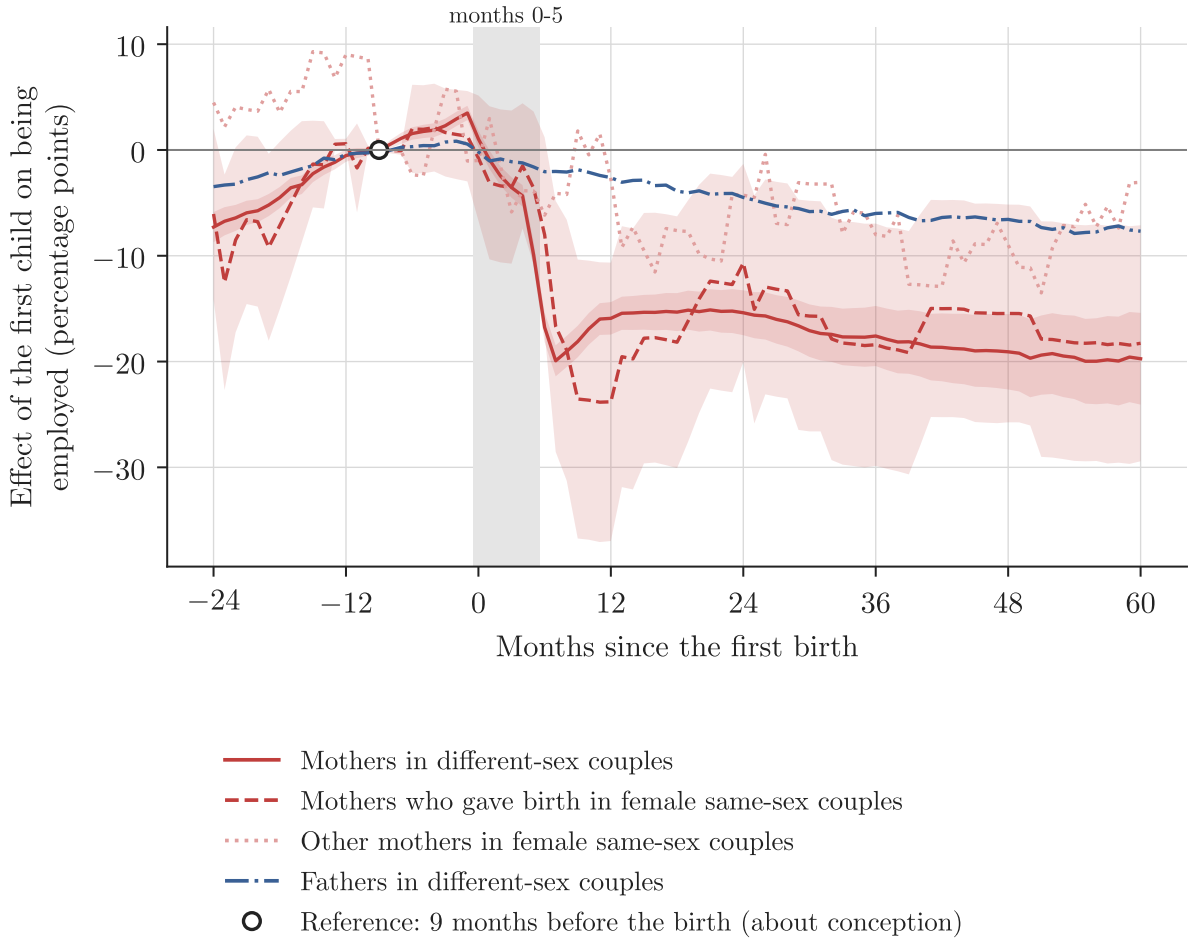


Figure 7: After the birth, the estimated employment loss of mothers who gave birth in female same-sex couples is close to that of mothers in different-sex couples.

Notes: Effect of the first child on being employed (positive employee contribution base in the month), in percentage points relative to month −9, by type of parent: event-study coefficients with parent-type fixed effects and age and year-by-month effects common by sex, estimated jointly by ordinary least squares without weights (Andresen and Nix, 2022). Single first births in 2018–2020 in couples without another child in t or $t+1$, both partners aged 22 to 60 and linked to the Social Security records, in which the mother who gave birth is identified by her birth benefit (Section 5.2) and was employed in the month before the birth: 47 female same-sex and 4,690 different-sex couples. 95 percent intervals clustered by couple, shown for the two groups of mothers. In months 6 to 11 employment falls by 19.1 percentage points (95 percent interval 7.7 to 30.5) for mothers who gave birth in female same-sex couples, 17.8 for mothers in different-sex couples and 2.1 for fathers; with 47 couples, only differences of about 16 points between the two groups of mothers can be detected with 80 percent power. For the other mother in female same-sex couples the estimate is −1.9; measured from month −12 it is −11.0, in line with Andresen and Nix (2022), who find that outside the leave co-mothers reduce their participation as much as birth mothers. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records.

6 Conclusion

The first birth reorganizes paid work within Spanish couples, and asymmetrically. Where the father was the breadwinner before conception, the share of couples with the father as sole earner rises sharply; where the mother was, the share with the mother as sole earner rises far less, if at all, and, in the raw transitions, dual-earning couples among them still move more often to the father as sole earner. Up to a 60 percent share of the couple's labor income, the mother loses far more of her earnings than the father, whoever earned more before the birth. Specialization on prior earnings predicts that the gap closes where she earned more; above 60 percent the data cannot tell whether it narrows, since the difference is significant only once the pre-birth drift is extended. Gender norms fit the pattern better, although the design does not separate them from other differences between mothers and fathers. The pattern resembles the one Querejeta (2026) documents for Uruguayan couples.

Two further pieces of evidence do not settle the question. Across Spanish regions, the asymmetry is not significantly larger where more people agree that a man's job is to earn money, and a single cross-section of regions cannot separate norms from everything else that differs between them. In female same-sex couples, the point estimates for the mother who gave birth, in income (-12.5 log points, against -13.2) and in employment (-19.1 percentage points, against -17.8), are close to those of mothers in different-sex couples, which is compatible with the birth, not prior labor income, driving the gap; with 27 and 47 couples, however, the income gap cannot be told from zero, nor either estimate from that of mothers in different-sex couples or from the half-size gap found in Norway.

In the year of the birth, the mother's own birth and childcare benefit, which replaces her pay during the leave, is about as large as the fall in her annual labor income, an accounting change rather than an effect of the birth. The extensions of paternity leave did not detectably change the child penalty in Spain (González et al., 2026), and here the fall in dual earning does not differ detectably across the four leave regimes ($p = 0.97$), a comparison that cannot rule out differences of up to about 9 points or separate the leave from the period of the birth.

The estimates center on months 6 to 11 after the birth and on couples of employees. Later waves of the panel will show whether the asymmetry persists as children grow, and a comparison with childless couples on the same monthly outcomes is left for future work.

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A Sample

Table 1 lists the couples left after each sample rule of the monthly analysis (Section 2), and Table 2 describes its main sample before the birth. Table 3 describes, in the year before the first birth, the different-sex, female same-sex and male same-sex couples of the comparison by prior labor income in Section 5.2; the male same-sex couples, too few to estimate the gap reliably, are shown for comparison.

	Couples
First child of both partners, a mother and a father, born 2017–2022	46,153
Both partners linked to the MCVL	45,677
Both aged 16–49 at the birth	45,114
Both in the same panel household in the birth year	27,794
Neither self-employed nor without an employee base, months −36 to −1	17,088
Neither with a wage that the Social Security records miss	16,217
Both with a Social Security record, months −60 to −25	10,605
Born 2017–2020, observed in months −24 to +60 (main sample)	7,044

Table 1: Sample selection.

Notes: Couples left after each rule, applied in order. The kinship records name both parents of 34 to 47 percent of the panel's children born in each year of 2017–2022. Months are counted from the birth, month 0. Without an employee base: days as an employee with no base in the records of employee contribution bases, which leave out agricultural employees and record those of domestic employees apart, with the bases of the self-employed. Wage that the Social Security records miss: a wage in the employers' withholding returns in the year before the birth and no contribution base as an employee that year, or a wage of at least 3,000 euros in the birth year and no registration as a worker that year. Social Security record: a registration of any kind, an unemployment benefit or a contribution base. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records (MCVL).

		Mother's share, months 36 to 25	
	All couples	Less than one half	At least one half
Mother's age at the birth (years)	33.1	32.8	33.7
Father's age at the birth (years)	35.2	35.3	35.2
Mother born abroad (%)	17.0	18.1	15.0
Father born abroad (%)	14.1	13.7	14.6
Mother employed (%)	86.9	83.3	94.8
Father employed (%)	91.1	95.2	84.1
Mother's monthly labor income (€)	1,615	1,308	2,212
Father's monthly labor income (€)	2,072	2,407	1,464
Both employed (%)	80.4	80.5	81.1
Only the father employed (%)	10.7	14.7	3.0
Only the mother employed (%)	6.6	2.8	13.7
Neither employed (%)	2.4	2.0	2.2
Mother's share of couple labor income (%)	43.7	34.4	61.3
Observations (couples)	7,044	4,396	2,524

Table 2: Couples of the monthly analysis before the birth.

Notes: Main sample of the monthly analysis (Table 1), in months 24 to 13 before the birth. Mother's share in months 36 to 25: of the couple's labor income in those months before the birth; 124 couples with no labor income in those months are only in the first column. Employed: positive labor income in the month. Monthly labor income: the contribution base as an employee, with the cap imputed and zeros included, in euros of each month. Mother's share: in months with positive couple income. Weighted with the panel's household weight. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records (MCVL).

	Different-sex	Female same-sex	Male same-sex
Lower earner's age (years)	33.6	33.9	36.5
Higher earner's age (years)	34.7	35.0	36.0
Lower earner's labor income (€)	19,763	15,219	27,060
Higher earner's labor income (€)	37,569	25,967	45,001
Lower earner's income share (%)	38.8	37.9	40.3
Lower earner earns more at $t = -1$ (%)	23.7	24.1	25.0
Both with wages (%)	84.4	81.9	85.0
Married (%)	68.1	91.6	85.0
Ranked by tax records (%)	42.8	42.2	40.0
Mother is the lower earner (%)	63.2	n.a.	n.a.
Observations (couples)	9,426	83	20

Table 3: Couples in the comparison by prior labor income, in the year before the first birth.

Notes: Means at $t = -1$, the year before the first birth, for the couples of the comparison by prior labor income in Section 5.2: first births in 2017–2022 in which both partners have positive and different prior labor income, measured as below, and a row in the tax records in $t = -1$ (a blank row counts as zero) and in at least one year from $t = 0$. Lower and higher earner: partners ranked by their mean labor income in the tax records over $t = -4$ and $t = -3$, in the years in which both have a tax record, or, failing that, by their mean employee contribution base in those years (ranked by tax records: share of couples ranked this way); the ranking is not updated at $t = -1$. Age: year of the birth minus year of birth. Labor income: wages and positive self-employment income in the tax records, in euros of each year, not capped. Lower earner's income share: of the couple's labor income at $t = -1$, where positive. Both with wages: both partners with positive wages in the tax records at $t = -1$. Married: spouses in the kinship records of any year from 2016 to 2023. n.a.: not applicable. The comparison by birth benefit in Section 5.2 uses a different selection, single first births in 2018–2020 in which the mother who gave birth is identified by her birth benefit (27 female same-sex and 3,071 different-sex couples). Unweighted. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records.

B Replication of de Quinto, Hospido and Sanz (2021)

The Household Panel is new, so this appendix checks it against the main Spanish estimate of the child penalty, that of de Quinto et al. (2021) from Social Security records. Their design is applied to the panel's parents with a first child in 1994–2009: the first child comes from the panel's kinship records, and earnings from the Social Security records that the panel links to each parent. Ten years after the first birth the penalty is 0.262, against their 0.282, and its average over years five to ten, the figure they compare across countries, is 0.251, against their 0.25 (Table 4). As they caution, the separate paths of mothers and fathers rest on a stronger assumption than the penalty; here both fall further than theirs in years one to ten (ten years after the birth, fathers are 21 percent below their counterfactual, against about 5 in the original, and mothers 46, against about 33), so only the penalty is compared. The check therefore covers the panel's parents and its link to the Social Security records, not its tax records.

	de Quinto et al. (2021)	Replication
Earnings, ten years after the birth	0.282	0.262
Earnings, average of years five to ten	0.25	0.251
Days worked, ten years after the birth	0.263	0.271

Table 4: The panel's estimate of the long-run Spanish child penalty is close to the published one.

Notes: Child penalty: the effect of the first child on fathers minus its effect on mothers, as a fraction of mothers' predicted outcome without children, equation (3) of de Quinto et al. (2021), estimated with their design on the parents of the Household Panel with a first child in 1994–2009 (105,641 mothers and 112,047 fathers), with earnings and days worked from the Social Security records linked to each parent. The original reports no standard errors; at ten years, the replication's are 0.008 for earnings and 0.006 for days worked (approximate, clustered by person). In the year of the birth the penalty depends on which twelve months are taken as year zero: with the twelve months ending in the month of the birth, as in the rows above, it is 0.069, against their 0.114; with calendar years, 0.129, against their 0.125. Unweighted; with the panel's weights, 0.251, 0.249 and 0.248. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records (MCVL).

C Transitions between employment arrangements

Figure 8 groups couples by who was employed at conception and shows how they spend months 6 to 11 after the birth. Couples in which only the mother was employed remain so in 37 percent of those months (95 percent interval 28 to 47), against 33 percent (26 to 40) of months −15 to −10 for couples in that arrangement 24 months before the birth: the two shares are close and their intervals overlap widely, although no test of their difference is computed and a sizable change cannot be ruled out. What differs is where the others go: both partners are employed in 39 percent of those months, against 57 percent before the pregnancy.

D The couple's income around the first birth

Figure 9 splits the change in the couple's annual income relative to the year before the birth into the labor income and the birth and childcare benefit of each parent, the tax credit for working mothers and other public transfers; the components add up exactly to the net change.

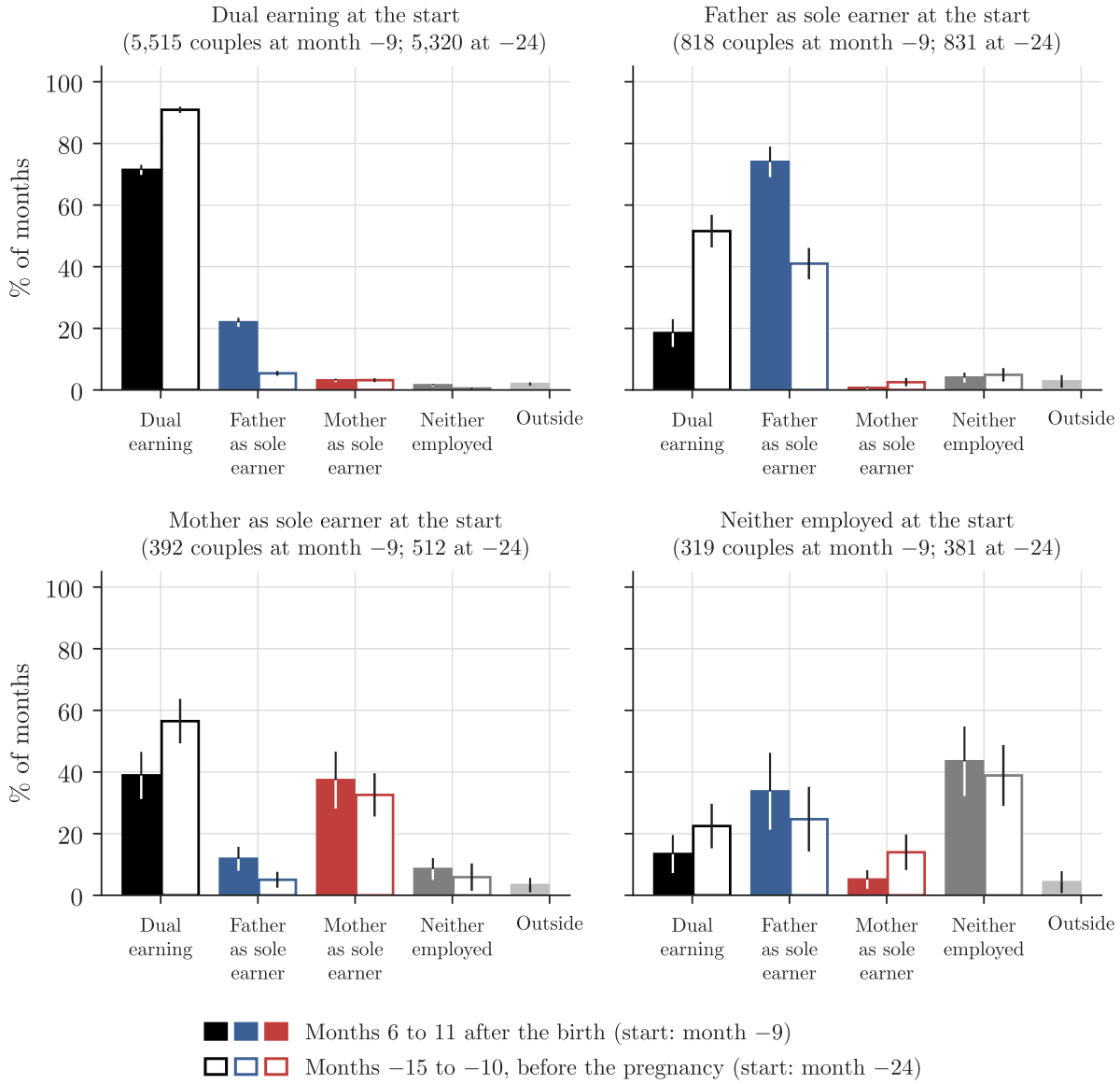


Figure 8: After the first birth, couples in which both were employed move far more often to the father as sole earner than to the mother as sole earner.

Notes: Filled bars: share of months 6 to 11 after the first birth spent in each arrangement, for couples grouped by their arrangement at month −9 (about conception). Hollow bars: share of months −15 to −10, before the pregnancy, for couples grouped by their arrangement at month −24. Employed: a positive Social Security contribution base as an employee in the month; birth leave counts as employed and unpaid childcare leave as not employed. Outside: months in which a partner is self-employed or an employee without a recorded contribution base; the sample excludes them before the birth, so this arrangement has no hollow bar. Vertical lines: 95 percent confidence intervals. Effective sample sizes with the weights at month −9: 99 couples with the mother as sole earner and 45 with neither employed. The hollow bars describe how arrangements changed before the pregnancy, not what would have happened without the child: they span 9 to 14 months instead of 15 to 20, and before conception couples were moving into dual earning (78.4 percent at month −24, 82.1 percent at months −15 to −10). Restricting to births in 2017–2018, whose months 6 to 11 precede March 2020, changes the shares from dual earning to a single earner, and from the mother as sole earner to dual earning or to the same arrangement, by at most 2.2 points. Sample: 7,044 couples with a first child born between January 2017 and December 2020, neither partner self-employed, in agriculture or in domestic service in the three years before the birth, Spanish Household Panel linked to Social Security records, weighted.

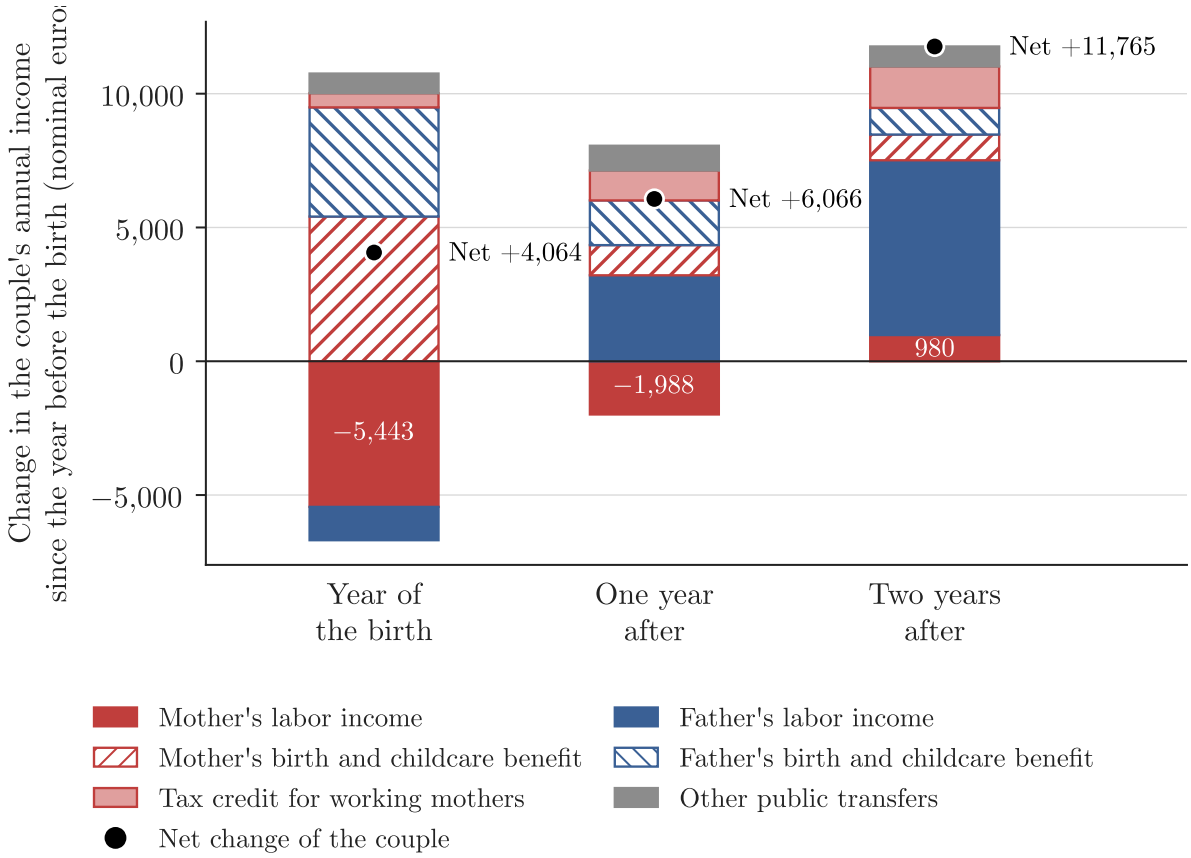


Figure 9: In the year of the birth, the mother's birth and childcare benefit, 5,408 euros, is as large as the fall in her labor income since the year before the birth, 5,443 euros.

Notes: Mean change in each component of the couple's annual income relative to $t = -1$, the year before the birth, in current euros of each year, where t is the calendar year minus the year of the first birth. Components that rise are stacked above zero and components that fall below it; the black point is the net change of the couple, the sum of all components. Labor income is wages plus self-employment income. Each birth and childcare benefit is the one paid to that parent. The tax credit for working mothers, paid almost entirely to mothers, includes its supplement for nursery costs. Other public transfers are all other public benefits of both parents in the income file: unemployment benefits, the pensions item and the remaining benefits. Labor income is capped at the 99th percentile of each parent's distribution. The mother's benefit at $t = 0$ is 99 percent of the fall in her labor income. These are accounting changes relative to $t = -1$, not effects of the birth: there is no counterfactual, so they also include the growth in income that would have occurred without a birth. Nominal euros, not deflated. Unweighted means over 3,013 different-sex couples with a first birth in 2019–2021 and positive joint labor income in every year from $t = -3$ to $t = +2$, from the annual tax records. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF).

E Earnings by who was the breadwinner

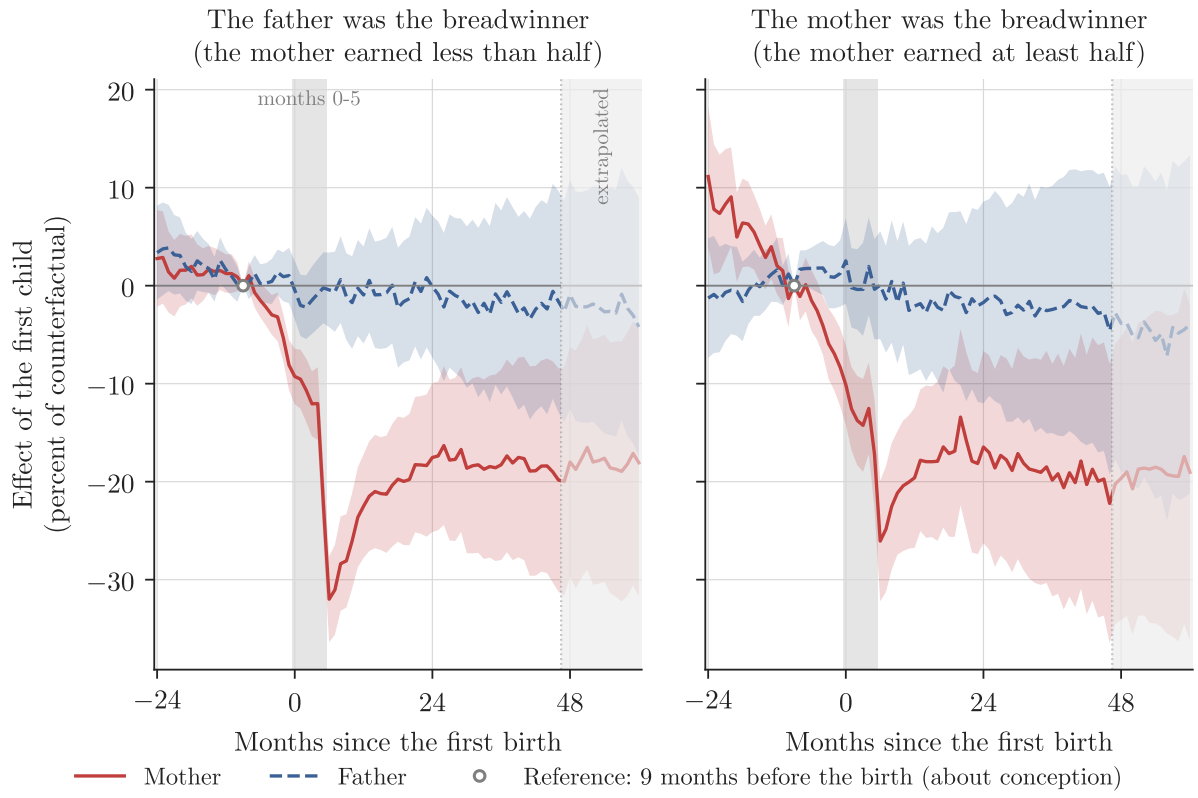


Figure 10: Six to eleven months after the first birth, the mother's labor income falls more than the father's, even when she was the breadwinner.

Notes: Each line is the estimated effect of the first child on that parent's monthly labor income, net of that parent's age and of year-by-month dummies, as a percentage of what the model predicts without the child, relative to 9 months before the birth (about conception; hollow circle). Employed: a positive Social Security contribution base as an employee in the month; monthly labor income is that base. Shading around each line: 95 percent confidence interval. Vertical grey band: months 0 to 5, when most mothers, but not all, take their 16-week maternity leave; during the leave the mother keeps her contribution base. Shaded and faded, from month 47 on: these months fall in December 2020 or later, when every couple in the sample already had its first child; with no couple still waiting for its child to compare with, the estimate is extrapolated and not interpreted. Panels: the breadwinner is the mother if she earned at least half of the couple's labor income 25 to 36 months before the birth, and the father otherwise (2,288 couples where the father was, 1,262 where the mother was). From 24 months before the birth to the reference month the mother's line goes from +11 percent to 0 in the right panel, and the other lines move by less than 4 percent; if each line's drift went on after the reference month, the gap between the father's and the mother's lines in months 6 to 11 would be 11 points instead of 22 in the right panel, and 29 instead of 28 in the left one. The right panel's gap minus the left one's is −6 points (95 percent interval −14 to 2), and −18 (95 percent interval −28 to −7) if the drifts went on. Without the child, in months 6 to 11, mothers would earn 1,949 and 3,154 euros a month and fathers 3,067 and 2,234. Sample: 3,550 couples with a first child born between January 2017 and December 2020 in which both partners were employed in each of those 12 months; neither self-employed, in agriculture or in domestic service in the three years before the birth; Spanish Household Panel linked to Social Security records; weighted. From the month of birth on, couple-months in which a partner is self-employed or an employee without a recorded contribution base are left out, not counted as zero: 1.6 percent of couple-months in months 6 to 11 and 6.9 percent from month 47 on. As in Machado and Jaspers (2023, figure 2), with the couples split at one half instead of three groups; unlike them, the coefficients on the age and year-by-month dummies are estimated within each panel. Before the reference month the joint test rejects that the mother's coefficients in the right panel are zero ($p = 0.04$), but not for the other three lines (p from 0.52 to 0.87).

F Robustness of the sole-earner asymmetry

Table 5 re-estimates the sole-earner asymmetry of months 6 to 11 under changes to the groups or the sample, each with the reference month, weights and controls of the main estimate, and also with the pre-birth trend extended as in Section 3. Comparing couples at the same distance from parity, the asymmetry is 16.4 points where the mother contributed 40 to 50 percent against 50 to 60 percent, 19.1 for 30 to 40 against 60 to 70 and 27.5 for 20 to 30 against 70 to 80; the first two survive extending the trend, and the third, with 303 couples on the mother's side, does not. Up to an 80 percent share, in every ten-point band of the mother's share the share with the father as sole earner rises, by 13 to 25 points, and the share with the mother as sole earner does not, before and after extending the trend; above 80 percent, with 558 couples, the bands are inconclusive once the trend is extended. Keeping only couples in which the same partner earned more in months -36 to -25 and in months -24 to -13, the asymmetry is 21.7, and keeping only couples in which both partners were employed in each of months -12 to -10, it is 19.5. It does not differ significantly across the four paternity leave regimes ($p = 0.13$), although under five and eight weeks it does not survive extending the trend. Among births up to March 2019, whose months 6 to 11 end before the COVID-19 pandemic, the asymmetry is 16.8, dual earning falls by 17.3 points and the mother's share by 12.3, against 17.9 and 10.7 in the full sample. After the birth, a couple-month is missing when a partner is self-employed or registered as an employee without a contribution base: 2.2 percent of couple-months in months 6 to 11, 0.8 percent of mothers' months and 1.5 percent of fathers', and 2.1 percent where the mother was the breadwinner against 2.2 where the father was. Setting every missing couple-month to the value that works most against the asymmetry still leaves 15.2 points.

G Gender norms across regions

Figure 11 maps the index by region, Table 6 reports the test of Section 5.1 and the same split for the mother's share of the couple's labor income, and Figure 12 the descriptive extension by thirds.

	Asymmetry	Trend extended	Couples
Main estimate	20.1 (16.2 to 24.0)	15.6 (9.1 to 22.1)	4,396 / 2,524
Same distance from parity, mother's share before conception			
40–50 against 50–60 percent	16.4 (12.0 to 20.8)	13.1 (5.9 to 20.3)	1,434 / 1,090
30–40 against 60–70 percent	19.1 (13.4 to 24.9)	16.7 (4.6 to 28.8)	1,060 / 573
20–30 against 70–80 percent	27.5 (18.7 to 36.4)	13.0 (−3.9 to 29.8)	760 / 303
Same breadwinner in two pre-birth years	21.7 (17.1 to 26.3)	15.2 (7.8 to 22.6)	3,855 / 1,934
Both employed in months −12 to −10	19.5 (16.6 to 22.5)	17.2 (11.8 to 22.7)	3,282 / 1,900
Paternity leave regime			
4 weeks	19.4 (7.8 to 31.0)	17.7 (5.1 to 30.2)	1,688 / 955
5 weeks	11.3 (−0.1 to 22.8)	1.1 (−15.2 to 17.3)	805 / 498
8 weeks	11.9 (0.9 to 23.0)	2.9 (−11.9 to 17.6)	808 / 441
12 weeks	21.6 (9.2 to 34.1)	32.6 (15.5 to 49.7)	1,095 / 630
Births up to March 2019	16.8 (9.4 to 24.2)	12.4 (3.6 to 21.1)	2,493 / 1,453
Missing months set against it	15.2 (11.2 to 19.2)	10.5 (3.9 to 17.0)	4,396 / 2,524

Table 5: The sole-earner asymmetry holds within 20 points of parity, with a persistent breadwinner, among prior dual earners, before the pandemic and with missing months set against it.

Notes: Sole-earner asymmetry in months 6 to 11 after the birth, in percentage points with 95 percent intervals: the rise in the share of couples with the father as sole earner among couples in which he was the breadwinner (first number of couples) minus the rise in the share with the mother as sole earner among couples in which she was (second number), relative to month −9, estimated as in Figure 3. Trend extended: the same difference after extending each group's pre-birth trend, a line through zero at the reference month fitted to the coefficients of months −24 to −10. Same distance from parity: the father-breadwinner group is the lower band of the mother's share in months −36 to −25 and the mother-breadwinner group the upper band. Same breadwinner in two pre-birth years: the same partner earned more in months −36 to −25 and −24 to −13. Paternity leave regime: event-time coefficients specific to each regime in one regression per group; a joint test does not reject that the four asymmetries are equal ($p = 0.13$). Births up to March 2019: month 11 falls in February 2020 at the latest. Missing months set against it: each couple-month in which a partner is self-employed or registered as an employee without a contribution base is set to 0 or 1, whichever lowers the asymmetry, or the trend-extended asymmetry in the second column, given that month's weight in the estimate. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records.

	Most traditional third	Other regions	Difference	Sign holds in three checks
Employment				
1. Father as sole earner, among couples where the father was the breadwinner	26.0 [20.5, 31.5]	16.1 [13.0, 19.2]	—	—
2. Mother as sole earner, among couples where the mother was the breadwinner	1.8 [−4.8, 8.5]	−2.6 [−5.4, 0.2]	—	—
3. Sole-earner asymmetry (row 1 minus row 2)	24.2 [15.5, 32.8]	18.7 [14.5, 22.9]	5.5 [−4.1, 15.1]	Yes
4. Dual earning, all couples	−23.0 [−27.9, −18.0]	−16.0 [−18.8, −13.3]	−6.9 [−12.6, −1.3]	Yes
Labor income				
5. Mother's share of the couple's labor income, all couples	−12.2 [−15.1, −9.3]	−10.2 [−11.7, −8.7]	−2.0 [−5.3, 1.3]	Yes

Table 6: In months 6 to 11 after the first birth, the sole-earner asymmetry is not significantly larger in the most traditional regions.

Notes: Effect of the first child in months 6 to 11 after the birth, in percentage points, relative to 9 months before the birth (about conception), from Equation (1) estimated by weighted least squares separately in each group; 95 percent intervals in brackets, with standard errors clustered by couple. Employed: a positive monthly contribution base; dual earning: both employed. The mother was the breadwinner if she earned at least half of the couple's labor income 25 to 36 months before the birth, and the father otherwise. Sole-earner asymmetry: the rise in the share of couples in which only the father is employed, among couples in which the father was the breadwinner, minus the rise in the share in which only the mother is employed, among couples in which the mother was the breadwinner; it is close to zero if couples reallocate paid work only according to who earned more. Row 5 is the mother's share of the couple's monthly labor income, added after the employment results; months in which the couple's labor income is zero or not fully recorded are left out. Regions grouped as in Section 5.1: the most traditional third of the 15 regions, listed under Figure 11, and the other ten. Difference: the most traditional third minus the other regions; the two groups share no couples, so their variances add. Gender norms predict a positive difference in row 3 and a negative one, a larger fall, in rows 4 and 5. The difference in rows 3 and 4 is the test of Section 5.1; rows 1 and 2 are both higher in the most traditional third, so their difference is not reported: row 3 nets out what raises both, if it acts equally on both. Last column: whether the difference keeps its sign with the pre-birth trend extended, with the three-item index and leaving out one region at a time; these give 11.0, 8.2 and 0.6 (without Galicia) to 7.4 in row 3, −5.3, −4.2 and −7.6 to −5.0 in row 4, and −3.5, −3.5 and −3.1 to −1.4 in row 5; of these, only the second in row 5 is significant at 5 percent ($p = 0.025$). In months −24 to −10, the coefficients of the other regions are jointly different from zero in rows 2 and 4 ($p = 0.024$ and $p = 0.035$). Couples, in the most traditional third and in the other regions: 1,026 and 3,287 in row 1, 562 and 1,900 in row 2, the two together in row 3, 1,626 and 5,270 in row 4, and 1,622 and 5,257 in row 5. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF), the linked Social Security records (MCVL) and CIS study 2942.

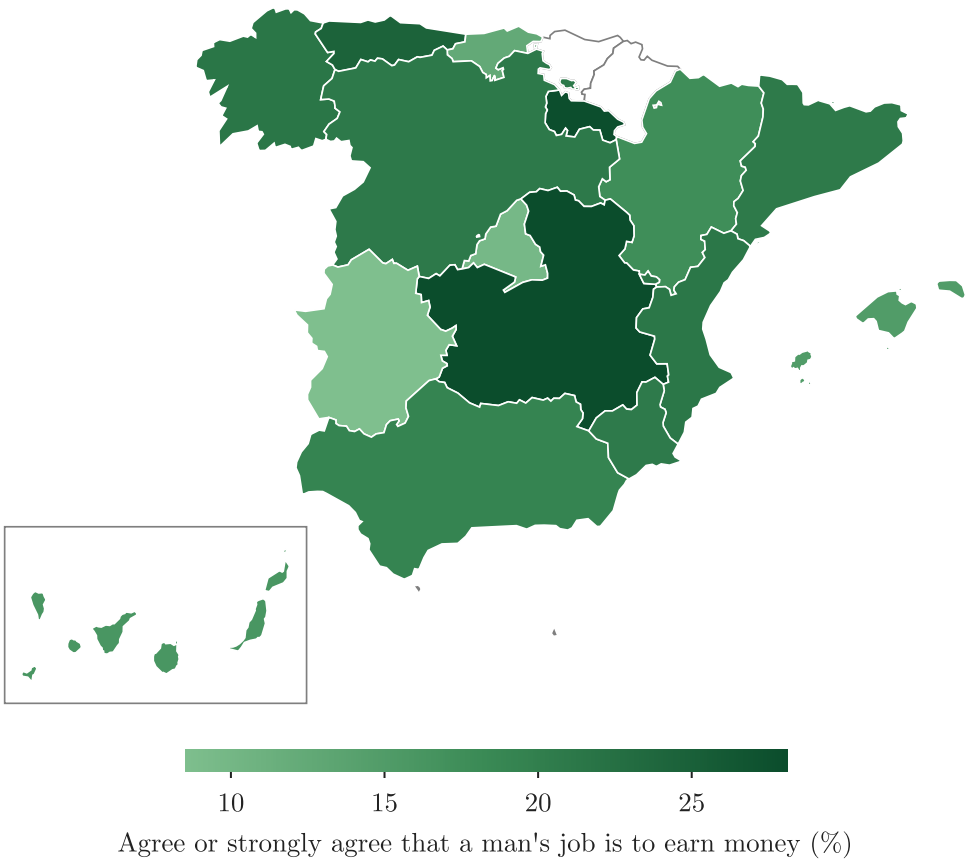


Figure 11: Agreement that a man's job is to earn money is about 10 percent in Madrid and 28 percent in Castile-La Mancha.

Notes: Share of respondents who agree or strongly agree that a man's job is to earn money and a woman's job is to look after the home and family, among those who answered, by region (ISSP 2012, Spanish fieldwork by the CIS, April 2012; 18 to 441 valid answers per region). The lowest share, 8.5 percent in Extremadura, rests on 59 answers (95 percent interval 1.4 to 15.6); Madrid's 9.7 rests on 350 (6.6 to 12.8). From light to dark green: from lower to higher agreement. Least traditional third: Balearic Islands, Canary Islands, Cantabria, Extremadura and Madrid; middle: Andalusia, Aragon, Castile and Leon, Catalonia and Murcia; most traditional: Asturias, Castile-La Mancha, Galicia, La Rioja and Valencia. Blank regions are not in this analysis: the Basque Country and Navarre are not in the panel, and Ceuta and Melilla have no interviews. The Canary Islands are moved closer to the mainland. Sources: CIS study 2942; Eurostat (GISCO), © EuroGeographics for the administrative boundaries.

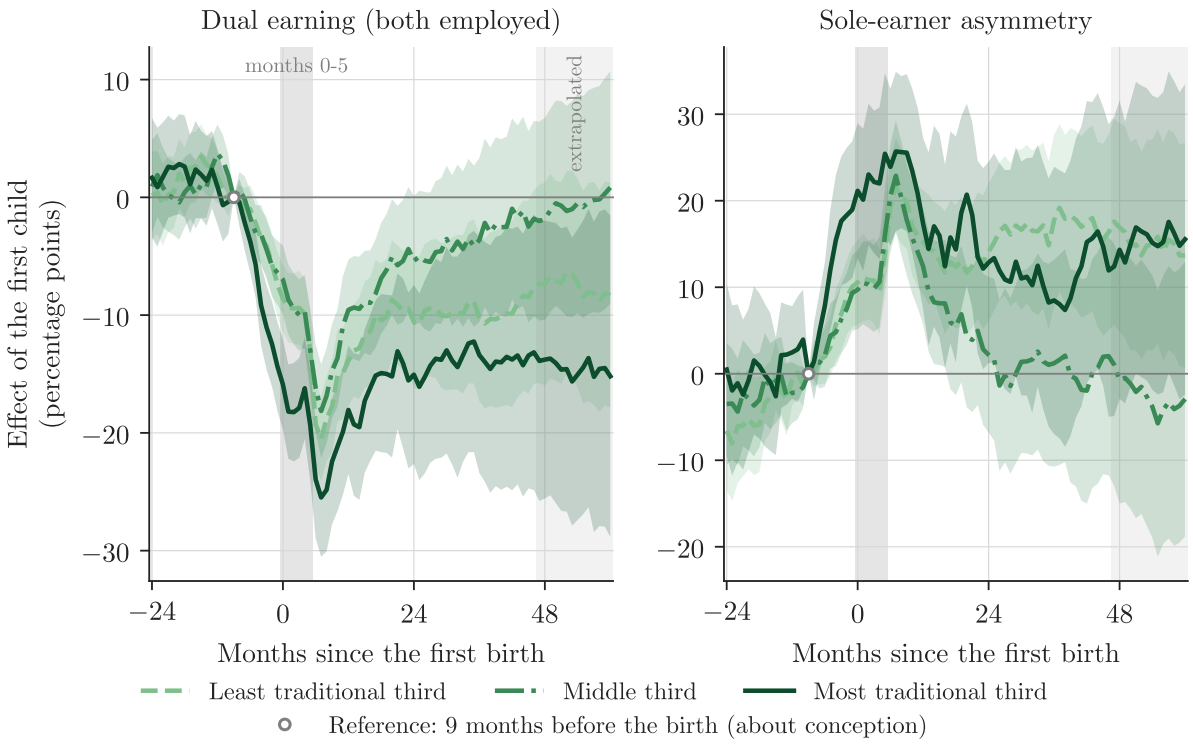


Figure 12: By thirds of regions, the effects of the first child do not grow steadily in size from the least to the most traditional third.

Notes: Lines: effects of the first child relative to 9 months before the birth (hollow circle), estimated by weighted least squares separately in each third of the 15 regions, grouped as in Figure 7 of Casarico and Lattanzio (2023) and listed under Figure 11; light, medium and dark green, least, middle and most traditional third. Left: dual earning (both employed); right: sole-earner asymmetry. Sole-earner asymmetry: the rise in the share of couples in which only the father is employed, among couples in which the father was the breadwinner, minus the rise in the share in which only the mother is employed, among couples in which the mother was the breadwinner; it is close to zero if couples reallocate paid work only according to who earned more. Bands: 95 percent confidence intervals. Grey band: months 0 to 5; light grey band, from month 47: extrapolated. In months −24 to −10, the coefficients are jointly different from zero in the middle third for dual earning ($p = 0.030$) and father as sole earner ($p = 0.008$), and in the least traditional third for mother as sole earner ($p = 0.040$). Descriptive, added after the test in Table 6; the thirds are not tested against each other. Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF), the linked Social Security records (MCVL) and CIS study 2942.

H Female same-sex couples

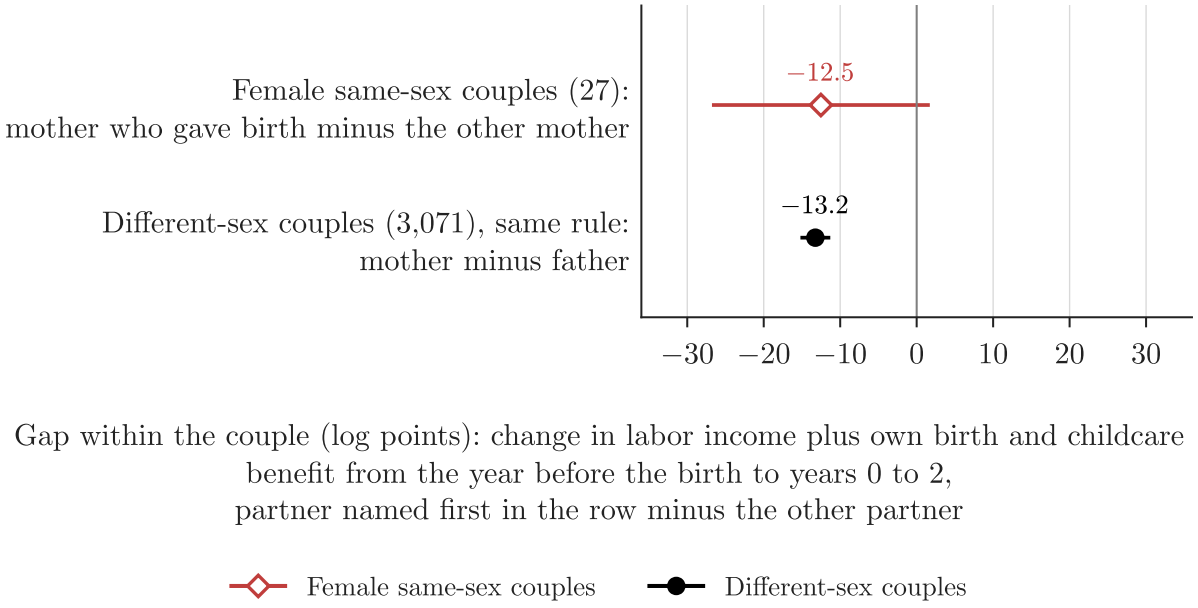


Figure 13: The point estimate for the mother who gave birth is close to that for mothers in different-sex couples, but 27 female same-sex couples cannot rule out no gap.

Notes: Change in labor income, plus own birth and childcare benefit, from the year before the birth ($t = -1$) to the year of the birth and the two following years ($t = 0$ to 2) of the partner named first in the row, minus the same change for the other partner, in log points, with 95 percent confidence intervals clustered by couple. Single first births in 2018–2020 in couples without another child in t or $t+1$. The mother who gave birth is the partner paid at least 15.5 weeks of birth benefit, measured with her contribution base of the month before the birth, when the other partner was paid fewer; for a single birth in these years the other parent could take at most 15 weeks, own leave plus any weeks transferred by the mother who gave birth (Farré et al., 2025; Real Decreto-ley 6/2019, 2019). The rule classifies only couples in which the mother who gave birth was an employee, or on unemployment benefit, in the month before the birth and took her full leave; in different-sex couples so classified it picks the mother in 98.7 percent of cases. Poisson pseudo-maximum likelihood with cohort fixed effects, the two partners in one regression, both in the tax records in $t = -1$ and in at least one year from $t = 0$; labor income in nominal euros, capped at the 99th percentile of each sample. By year, the female same-sex gap is +3.4 (s.e. 7.8) at $t = 0$, −42.6 (14.9) at $t = 1$ and +1.6 (12.0) at $t = 2$, against −7.9, −18.6 and −13.2 in different-sex couples; at $t = -2$ it is +0.6 (13.6), with 17 couples observed. With the partner paid more weeks of leave, measured with the salary of $t = -1$, standing in for the mother who gave birth, 44 couples give −10.6 (−33.6 to 12.3). Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records.

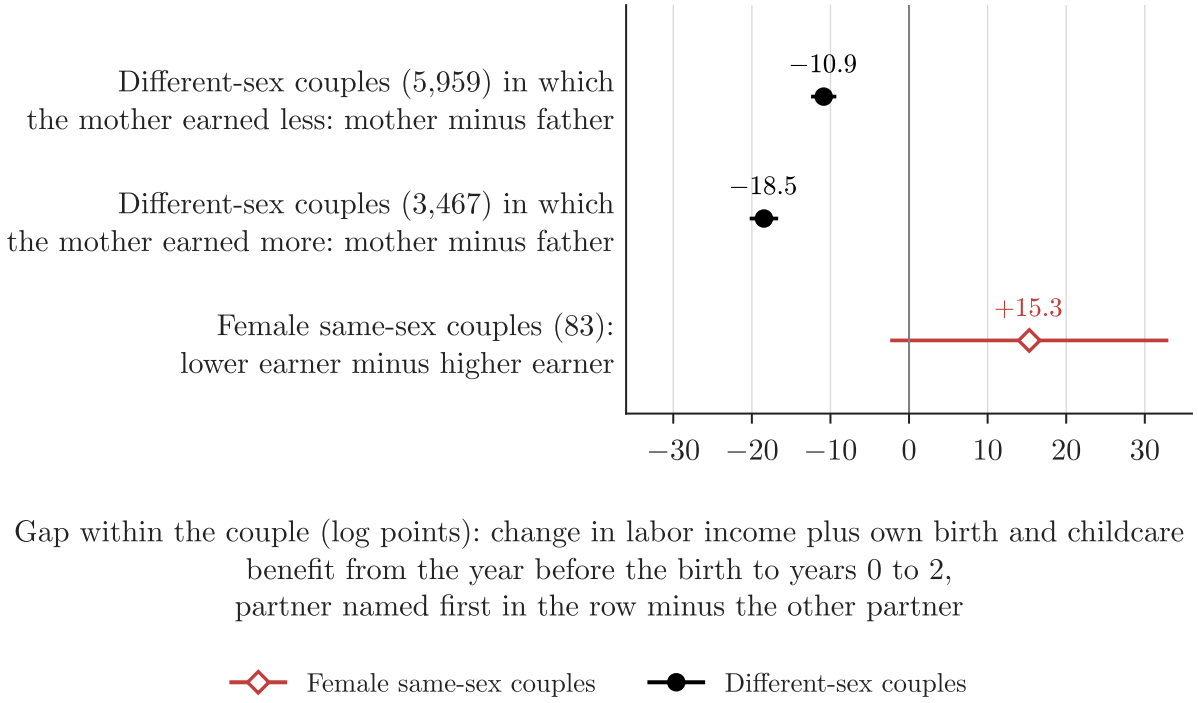


Figure 14: In different-sex couples the mother falls behind the father whether she earned less or more before the birth; in female same-sex couples the gap by prior labor income is imprecise.

Notes: Change in labor income, plus own birth and childcare benefit, from the year before the birth ($t = -1$) to the year of the birth and the two following years ($t = 0$ to 2) of the partner named first in the row, minus the same change for the other partner, in log points, with 95 percent confidence intervals clustered by couple. First births in 2017–2022; partners ranked by their mean labor income in the tax records over $t = -4$ and $t = -3$, in the years in which both have a tax record, or, failing that, by their mean employee contribution base in those years; births in 2022 are observed up to $t = +1$; for births in 2017, the birth benefit paid that year was taxable, is recorded with pensions in the tax records and is not included. Poisson pseudo-maximum likelihood with cohort fixed effects, the two partners in one regression, both in the tax records in $t = -1$ and in at least one year from $t = 0$; labor income capped at the 99th percentile of each sample. Labor income is in nominal euros: the price level is common to both partners and largely cancels in the gap, which moves by less than one log point when incomes are deflated by the consumer price index. Ranking on past labor income makes the lower earner's income grow faster with or without a birth. The female same-sex estimate is +37.4 (s.e. 14.5) among the 35 couples ranked by tax records and −2.7 (s.e. 10.7) among the 48 ranked by contribution bases, and between $t = -2$ and $t = -1$ it had already moved by +17.5 (s.e. 9.5). Source: Author's calculations based on the Panel de Hogares (INE, AEAT, IEF) and the linked Social Security records.